End-to-End Encrypted Collaborative Documents for Investigative Journalists

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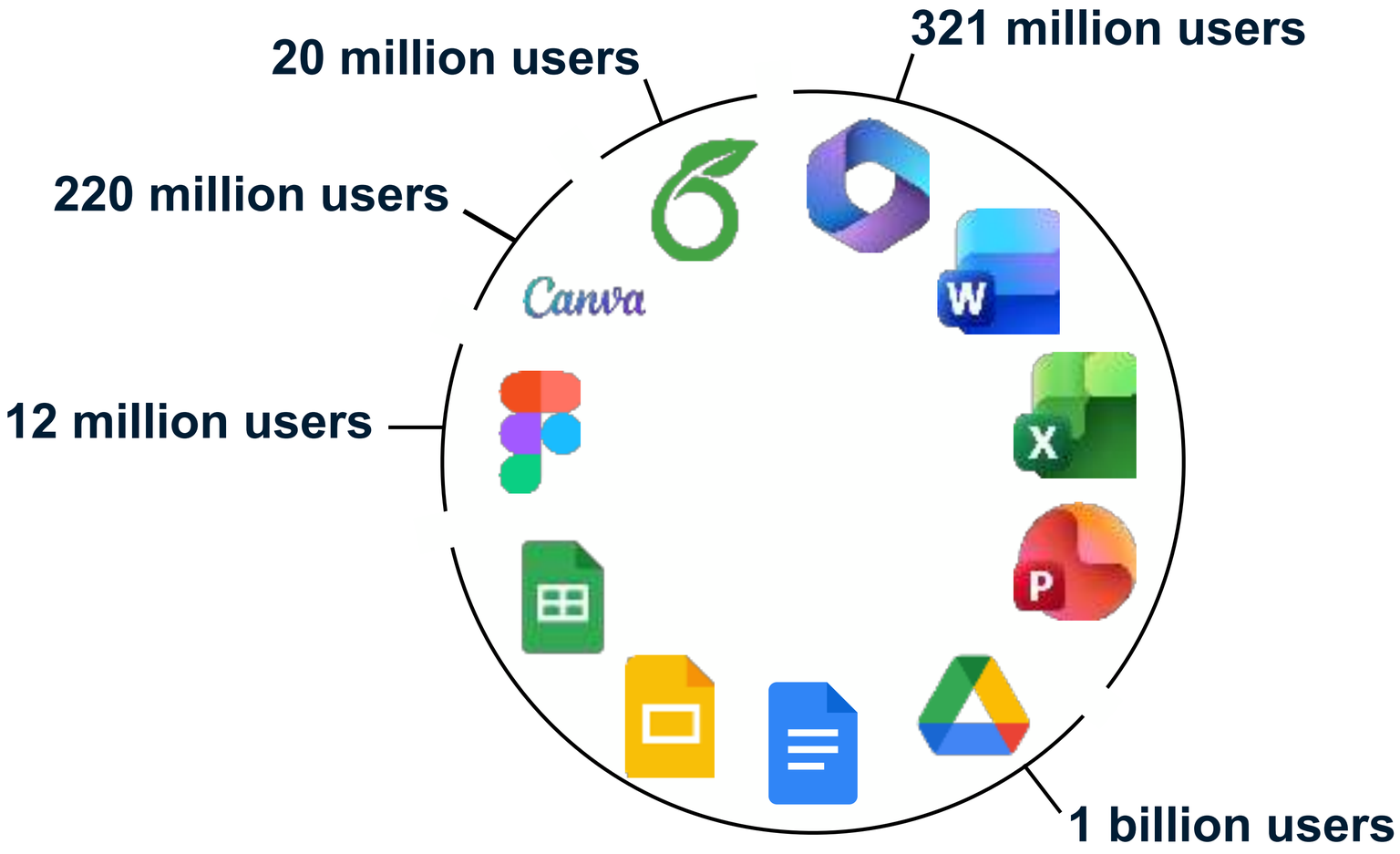
Zayd Maradni MPI-SP
Carmela Troncoso EPFL, MPI-SP

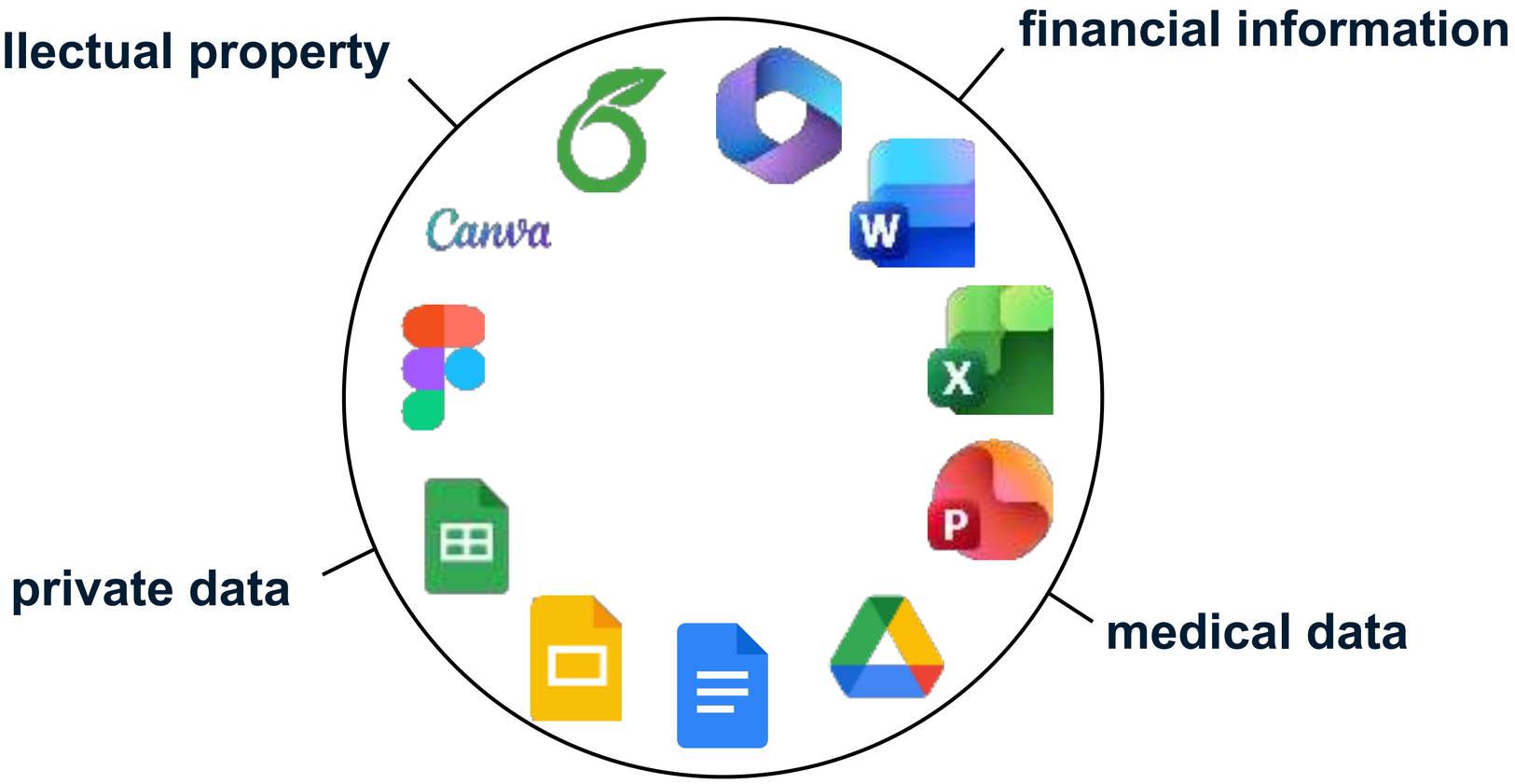
To appear at USENIX Security'26

Collaborative documents

Collaborative documents are ubiquitous



Collaborative documents are sensitive



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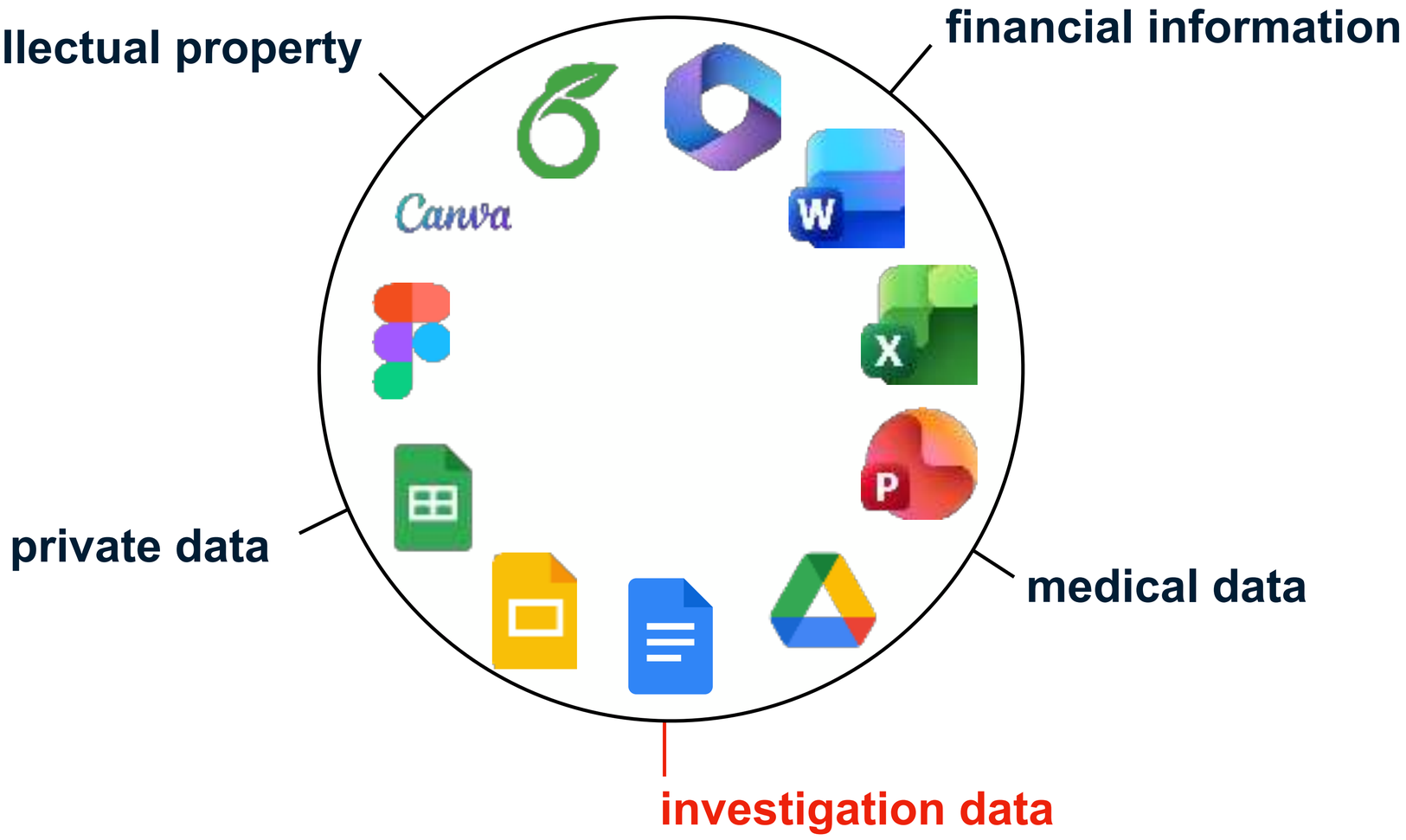

Requirements of secure collaborative documents for investigative journalists

Requirements

with thanks to INTERNATIONAL CONSORTIUM OF INVESTIGATIVE JOURNALISTS

Functional

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Sharing docs

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against malicious server

Existing secure collaborative docs

Existing secure collaborative docs

	satisfies functional reqs.	satisfies deployment reqs.	(provably) satisfies security reqs.
Academic Capsule [Kob18] SnapDoc [KKB19] HE/MPC [BFMPP21,PPJP23]	?	✗	✔

[Kob18] Kobeissi, “Capsule: A Protocol for Secure Collaborative Document Editing”, ePrint’18
[BFMPP21] Barbosa, Ferreira, Marques, Portela, Preguiça, “Secure conflict-free replicated data types”, ICDCN’21
[PPJP23] Portela, Pacheco, Jorge, Pontes, “General-purpose secure conflict-free replicated data types”, CSF’23
[KKB19] Kollmann, Kleppmann, Beresford, “Snapdoc: Authenticated snapshots with history privacy in peer-to-peer collaborative editing”, PETS’19

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Also: E2EE cloud storage, E2EE Git

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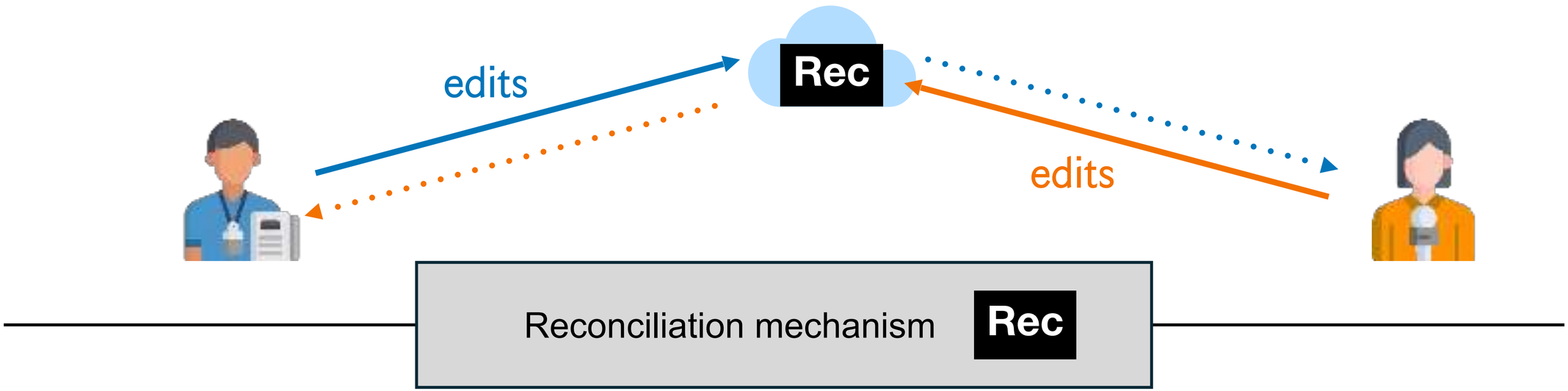
Building secure collaborative documents

Collaborative documents

Reconciliation mechanism **Rec**

Collaborative documents



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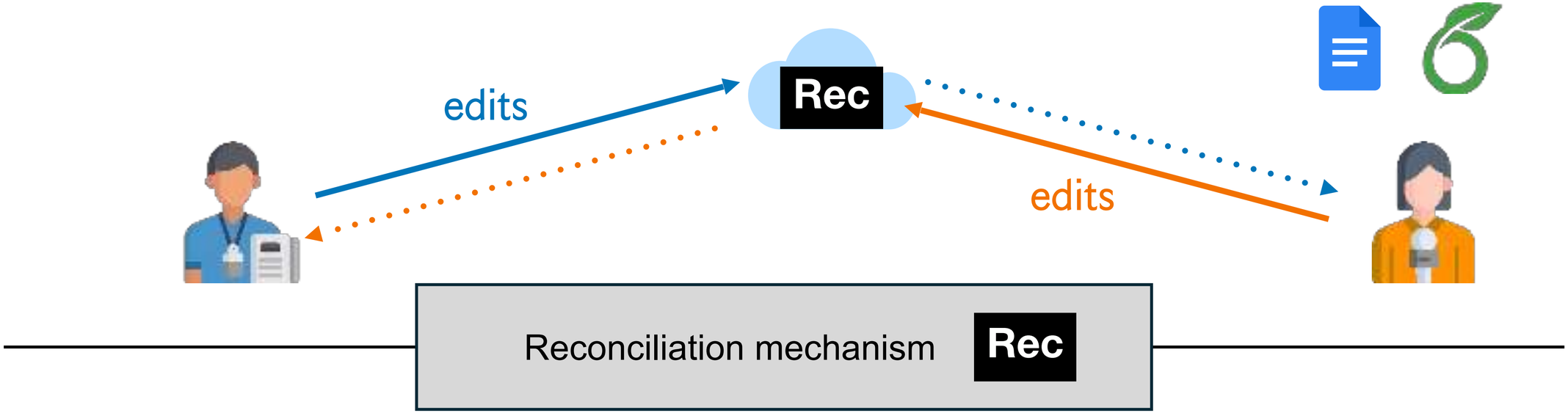

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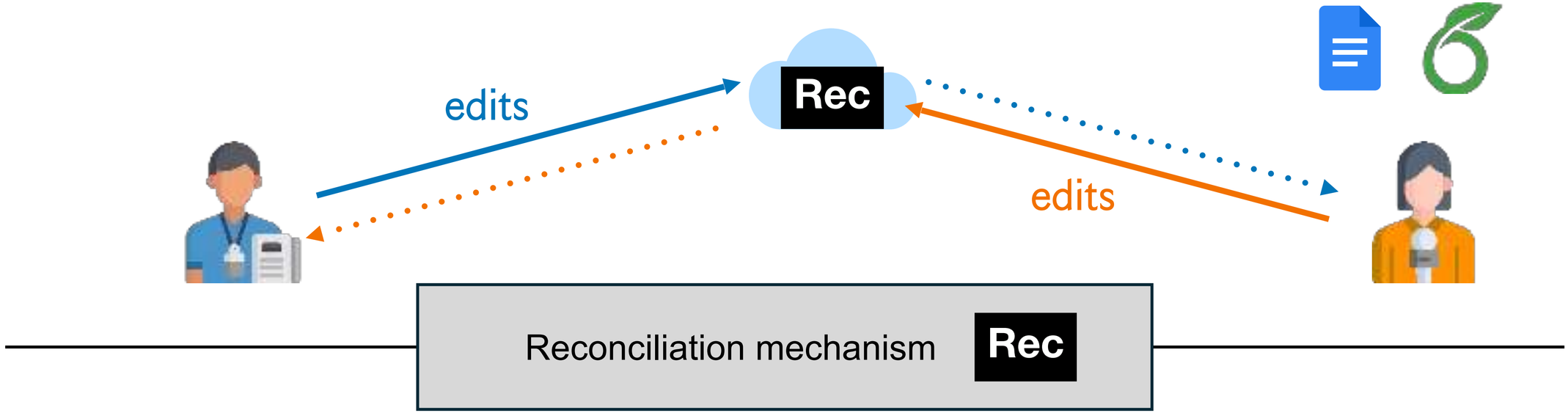

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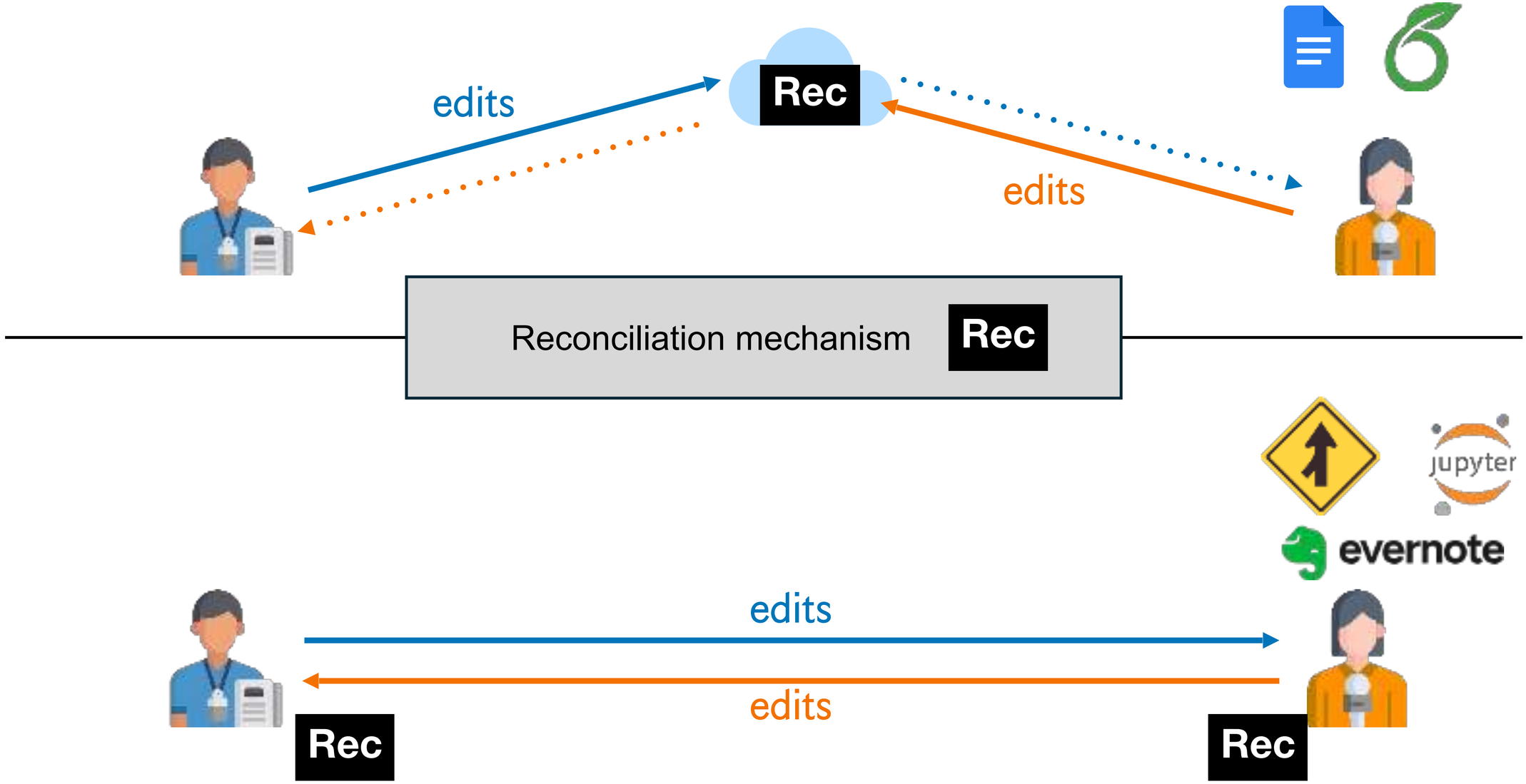

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Users receiving the same set of edits, regardless in which order, have the same view of the document

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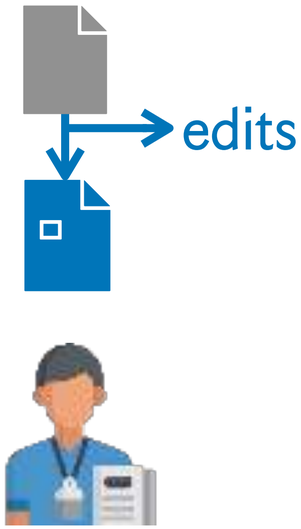

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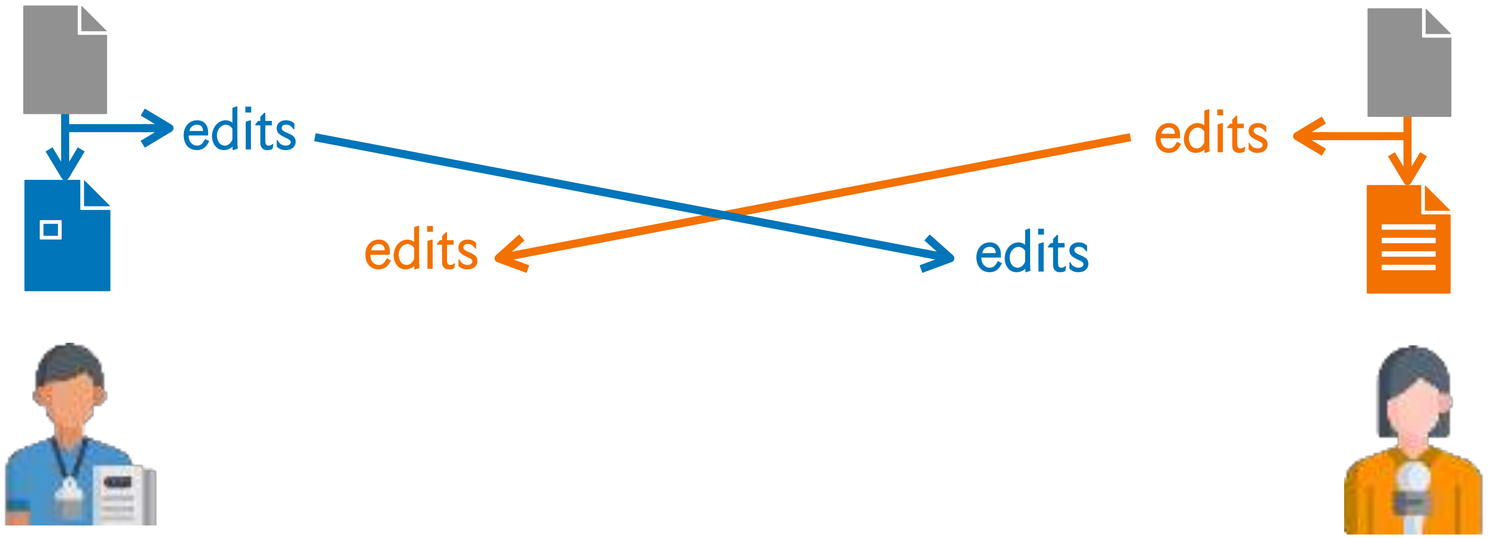

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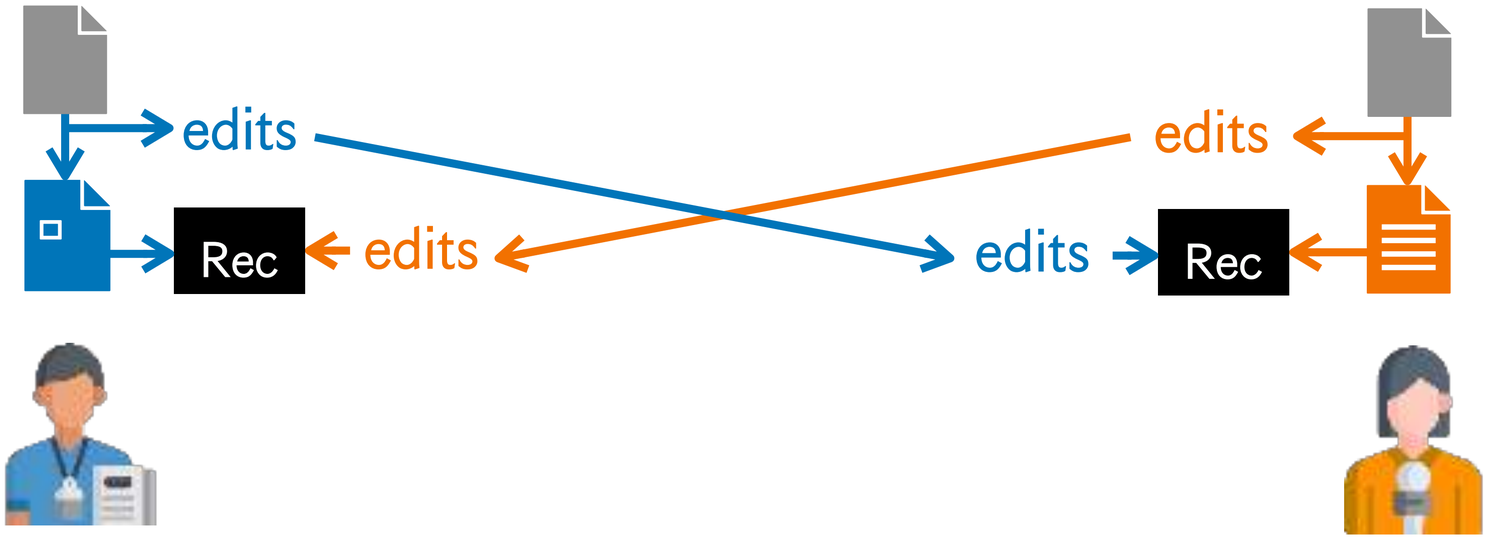

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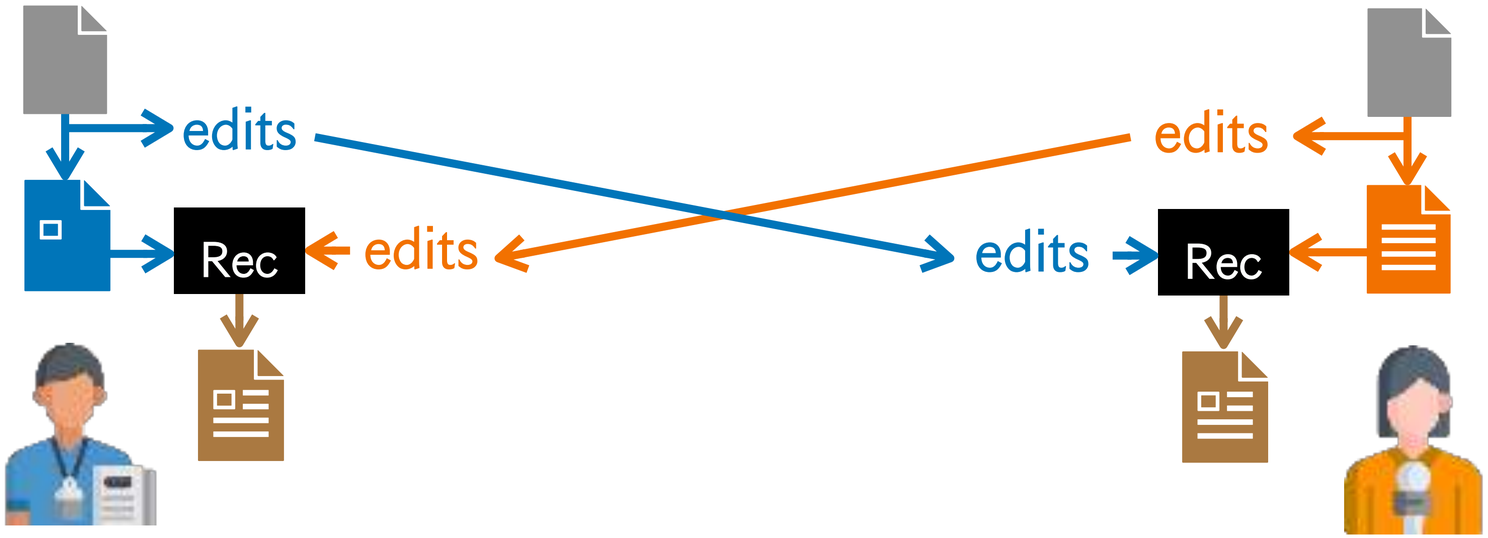
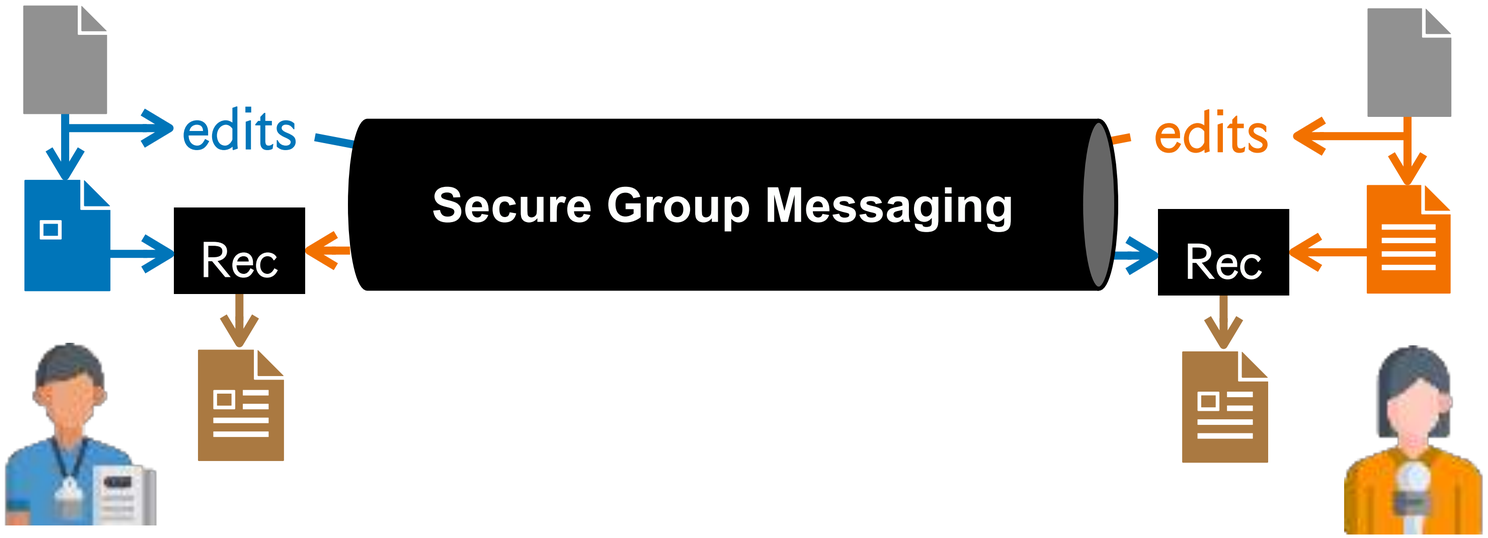


Secure Group Messaging is sufficient for E2EE CD



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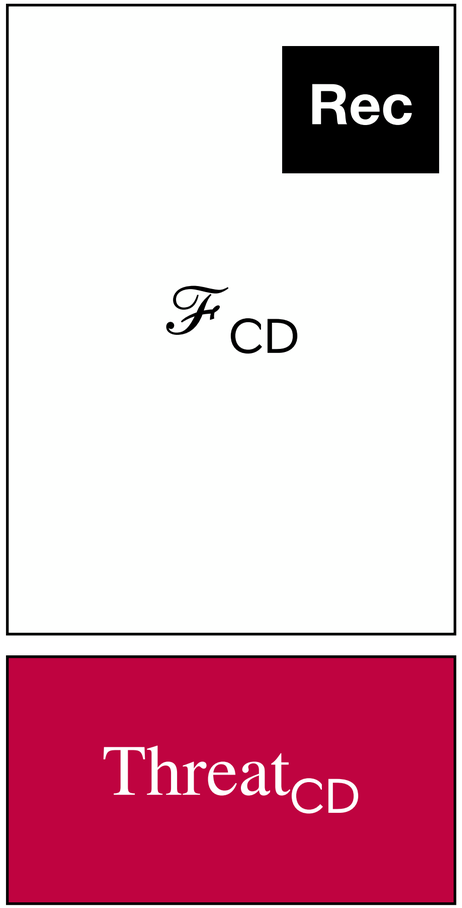
Formal



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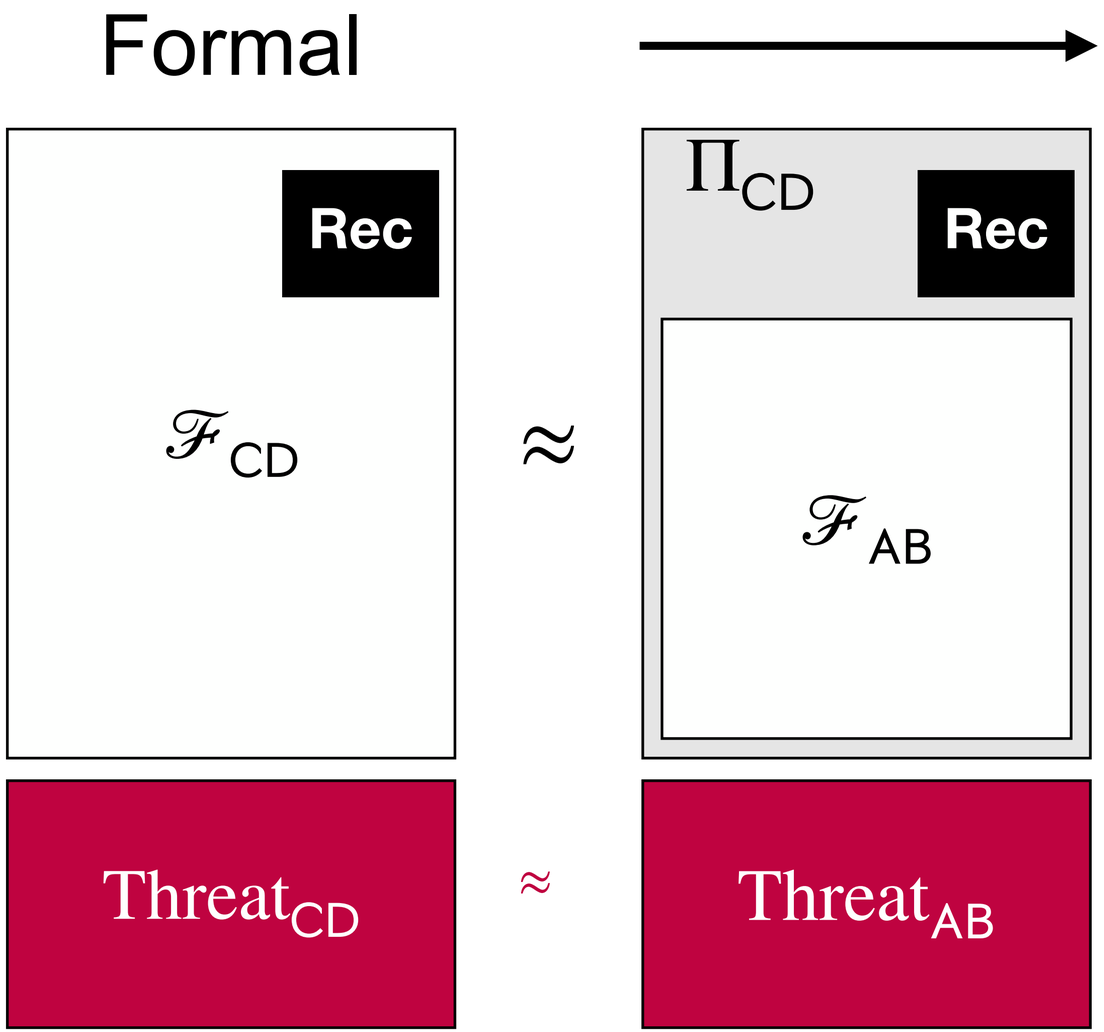

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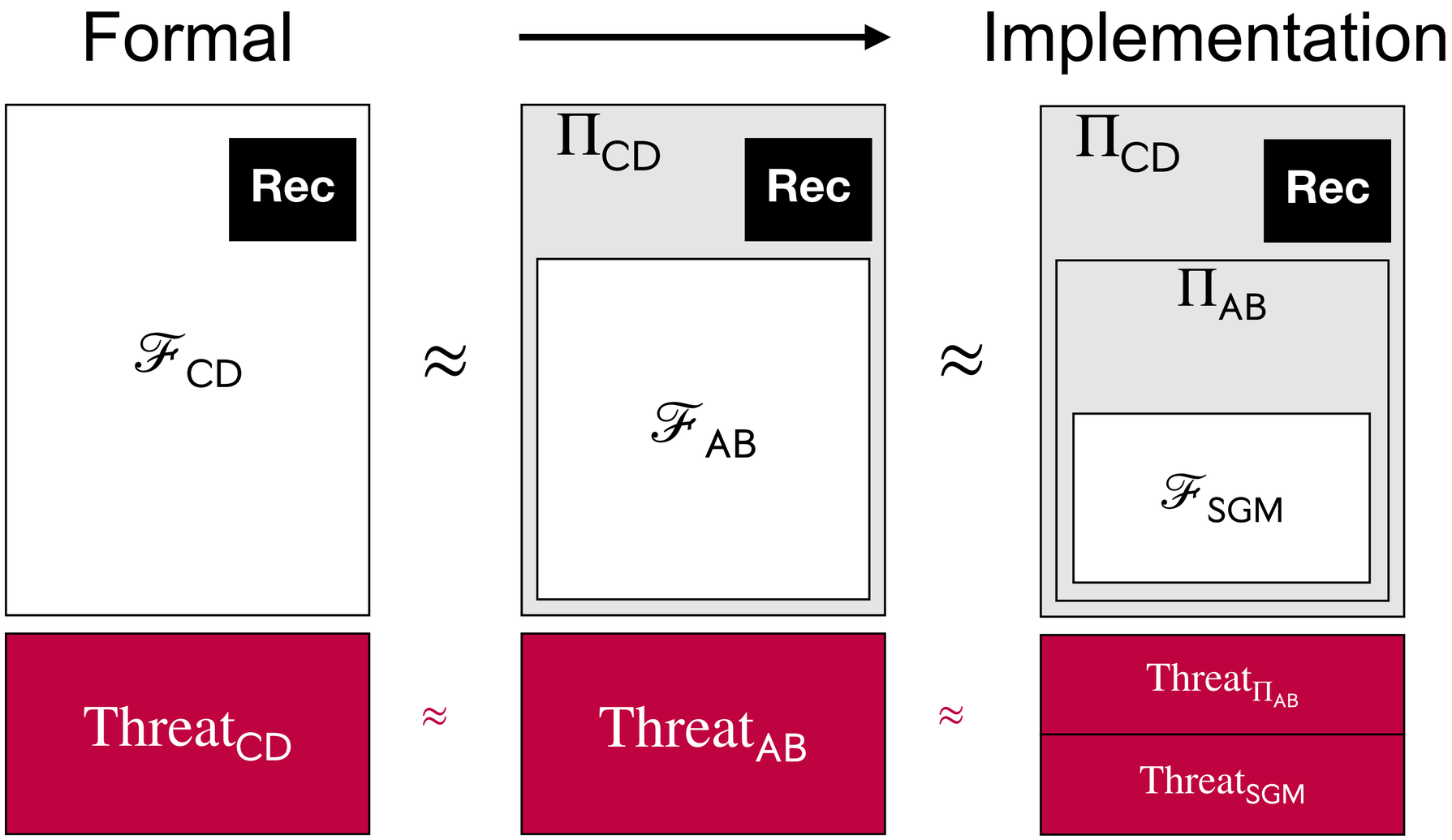


E2EE CD

Setup

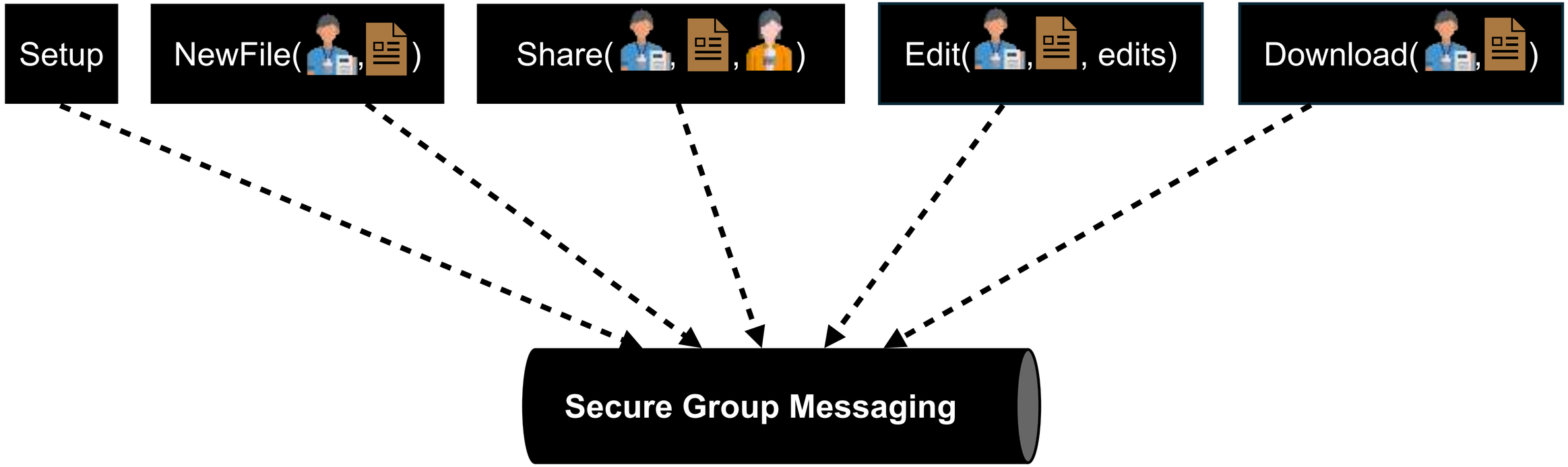
NewFile(,)

Share(, ,)

Edit(, , edits)

Download(,)

E2EE CD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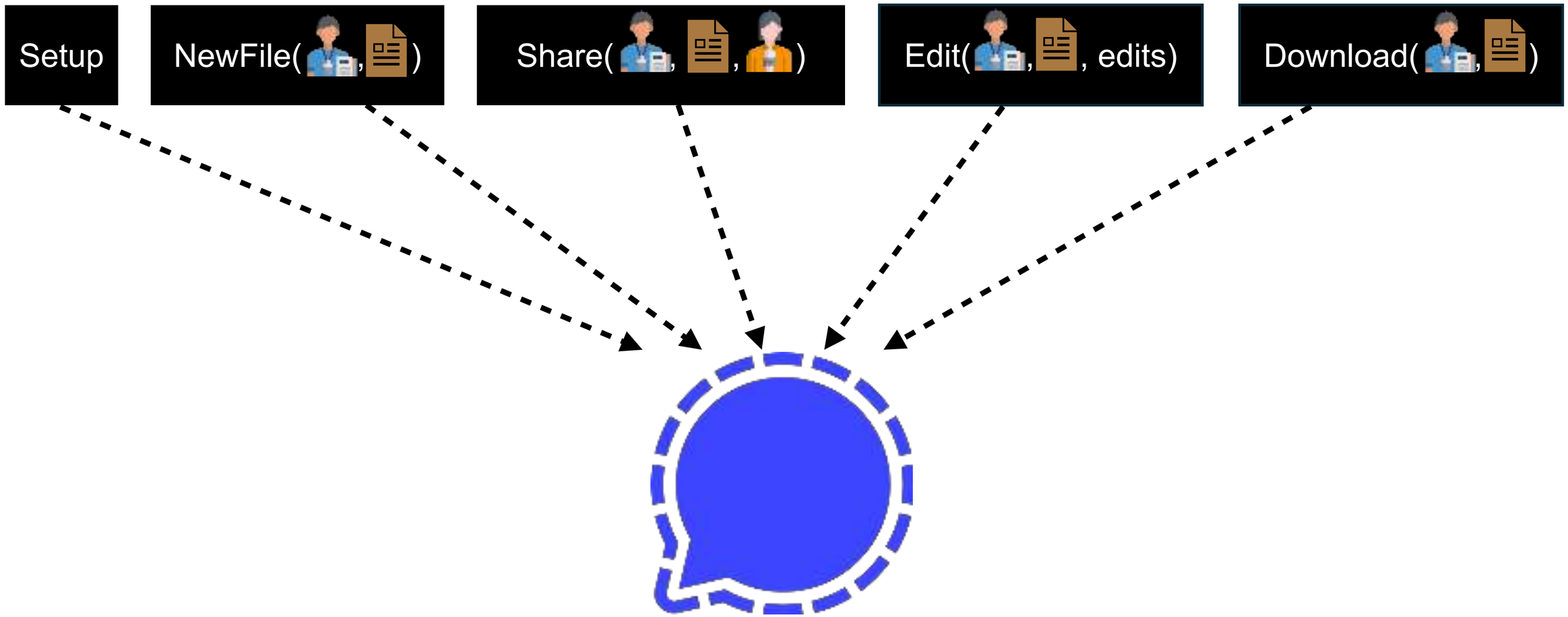

E2EE CD from Signal

Setup

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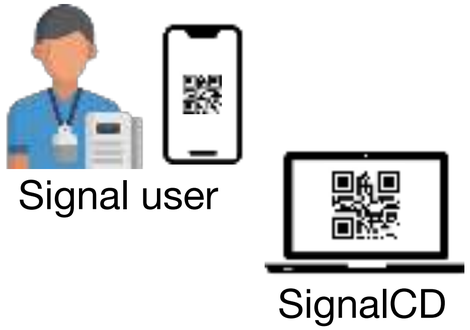
Setup

Setup: link account



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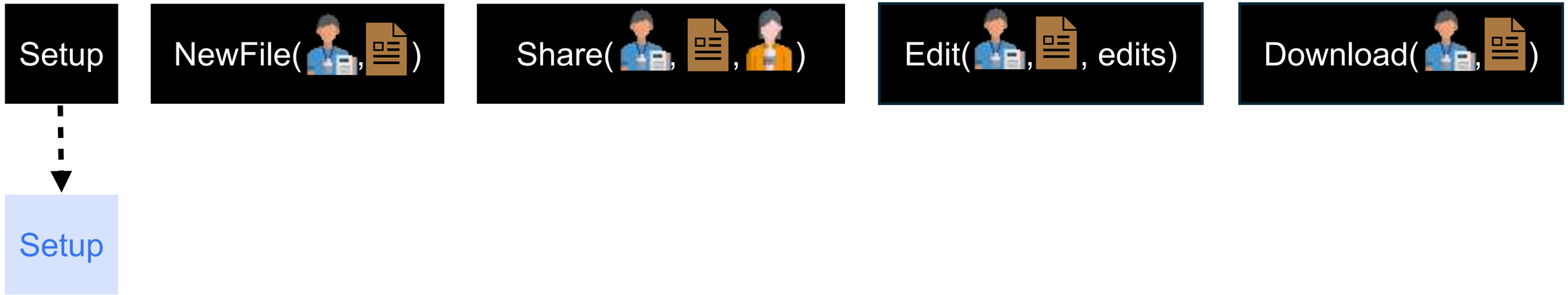

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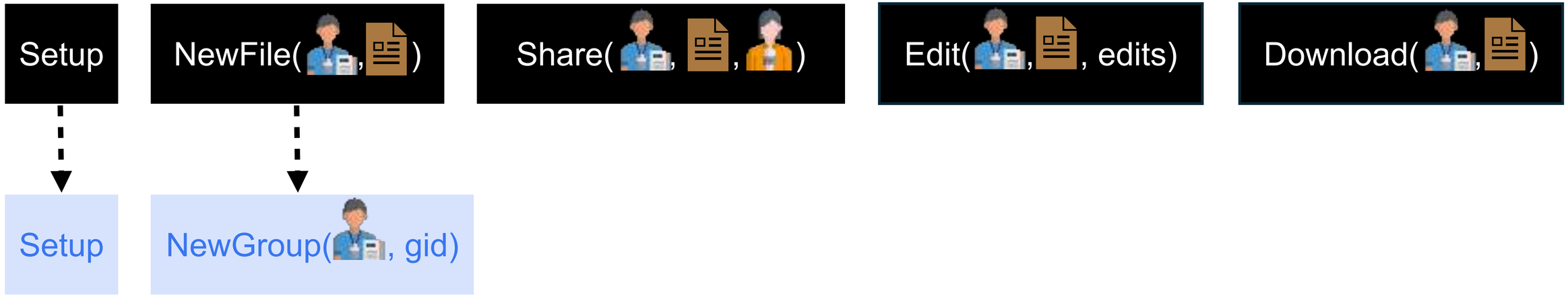

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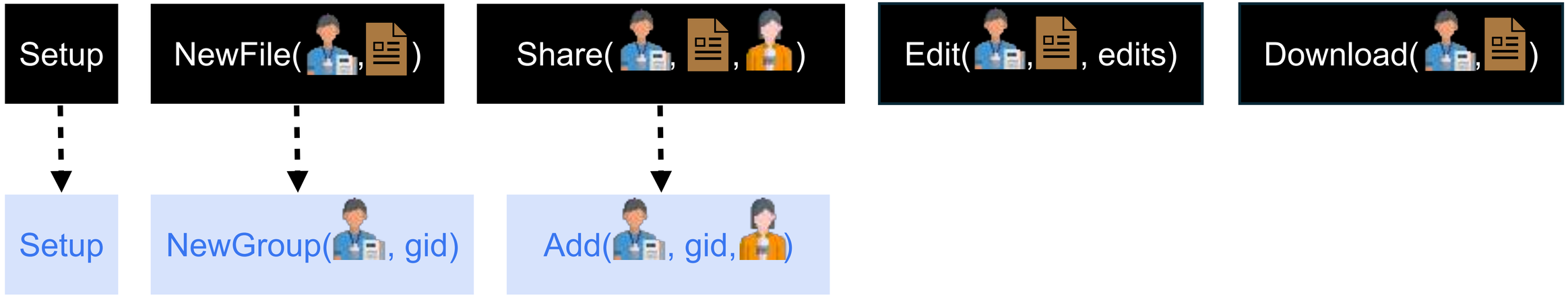

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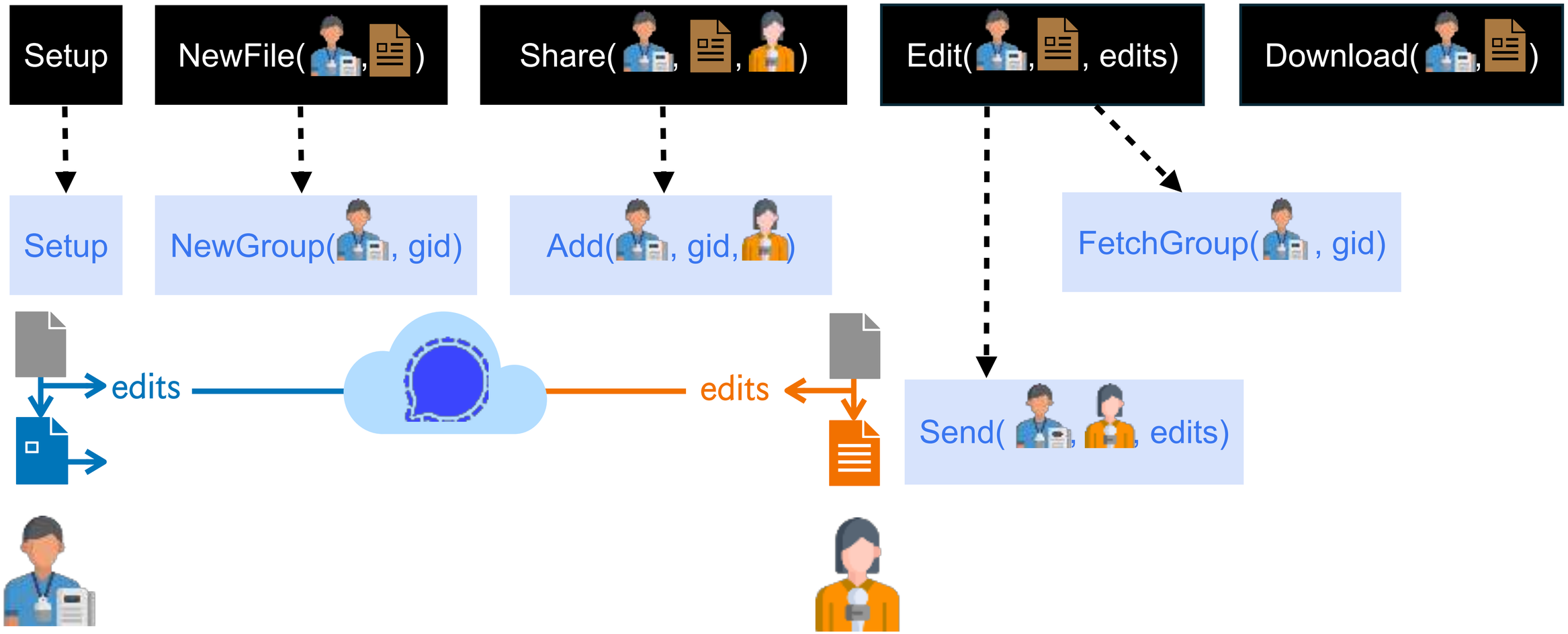

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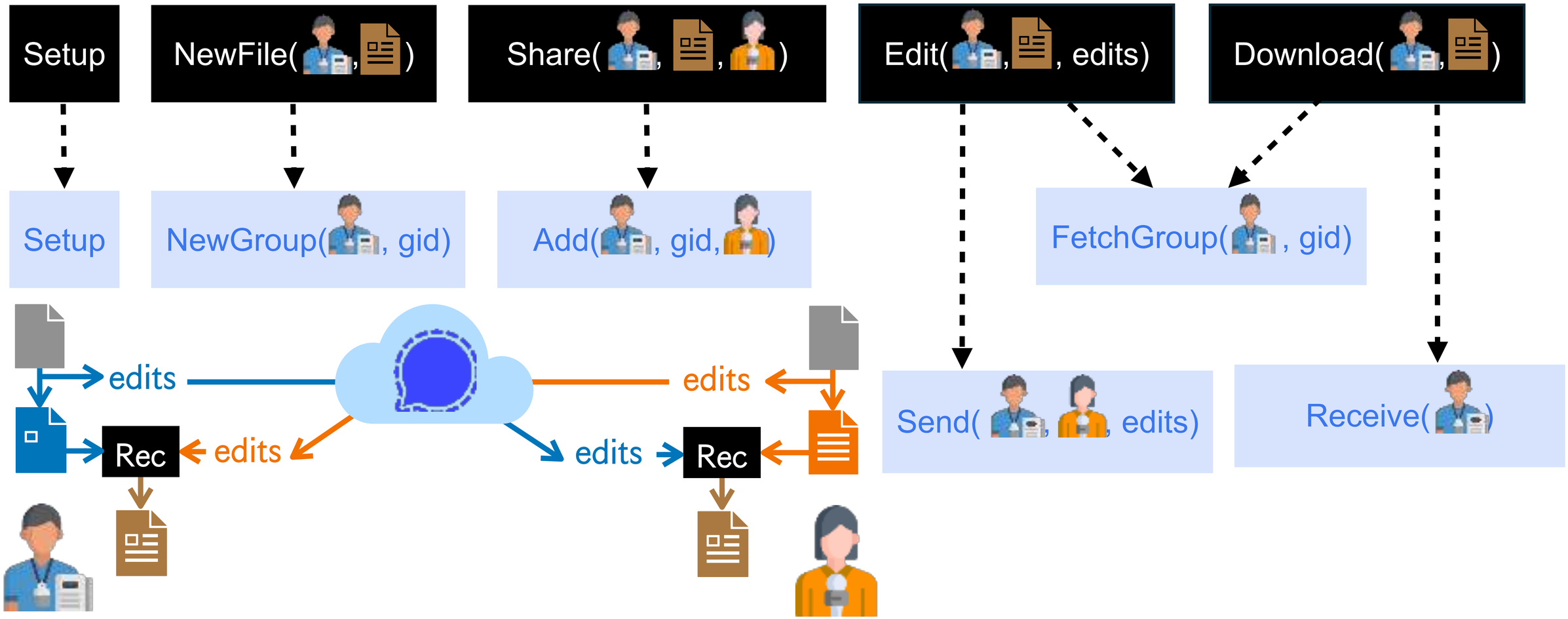

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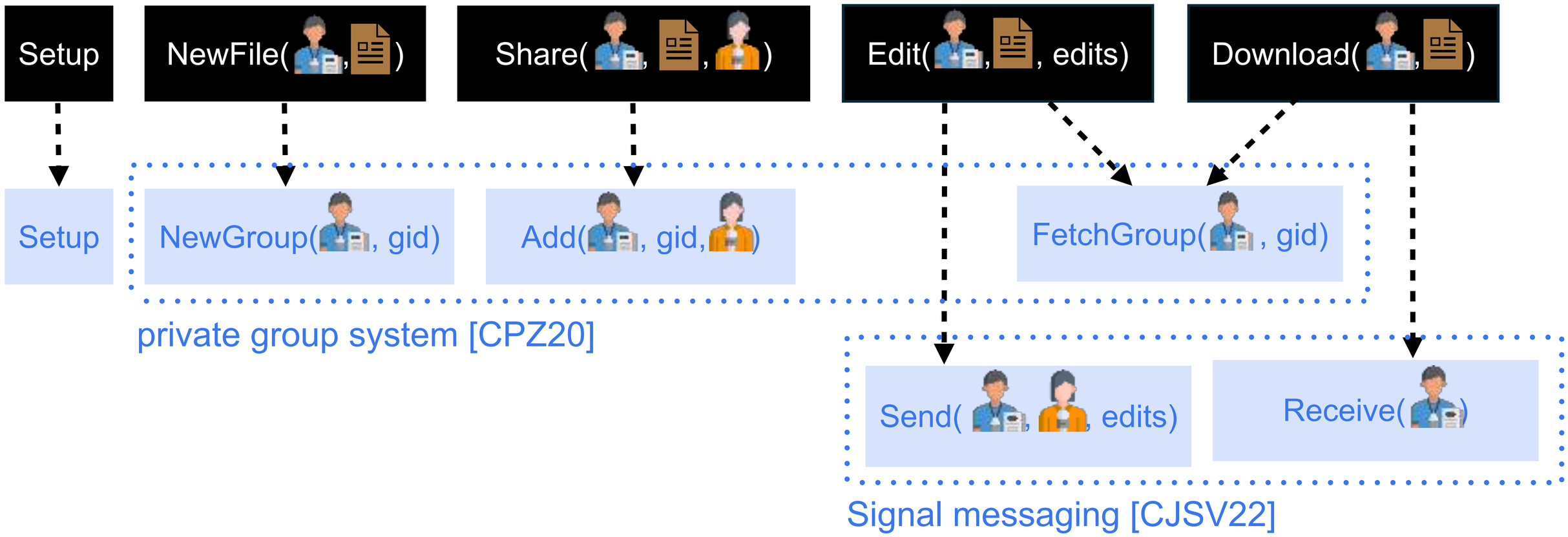

[CPZ20] Chase, Perrin, Zaverucha, “The Signal Private Group System and Anonymous Credentials Supporting Efficient Verifiable Encryption”, CCS’20
[CJSV22] Canetti, Jain, Swanberg, and Varia, “Universally Composable End-to-End Secure Messaging”, CRYPTO’22

Evaluation

Baseline

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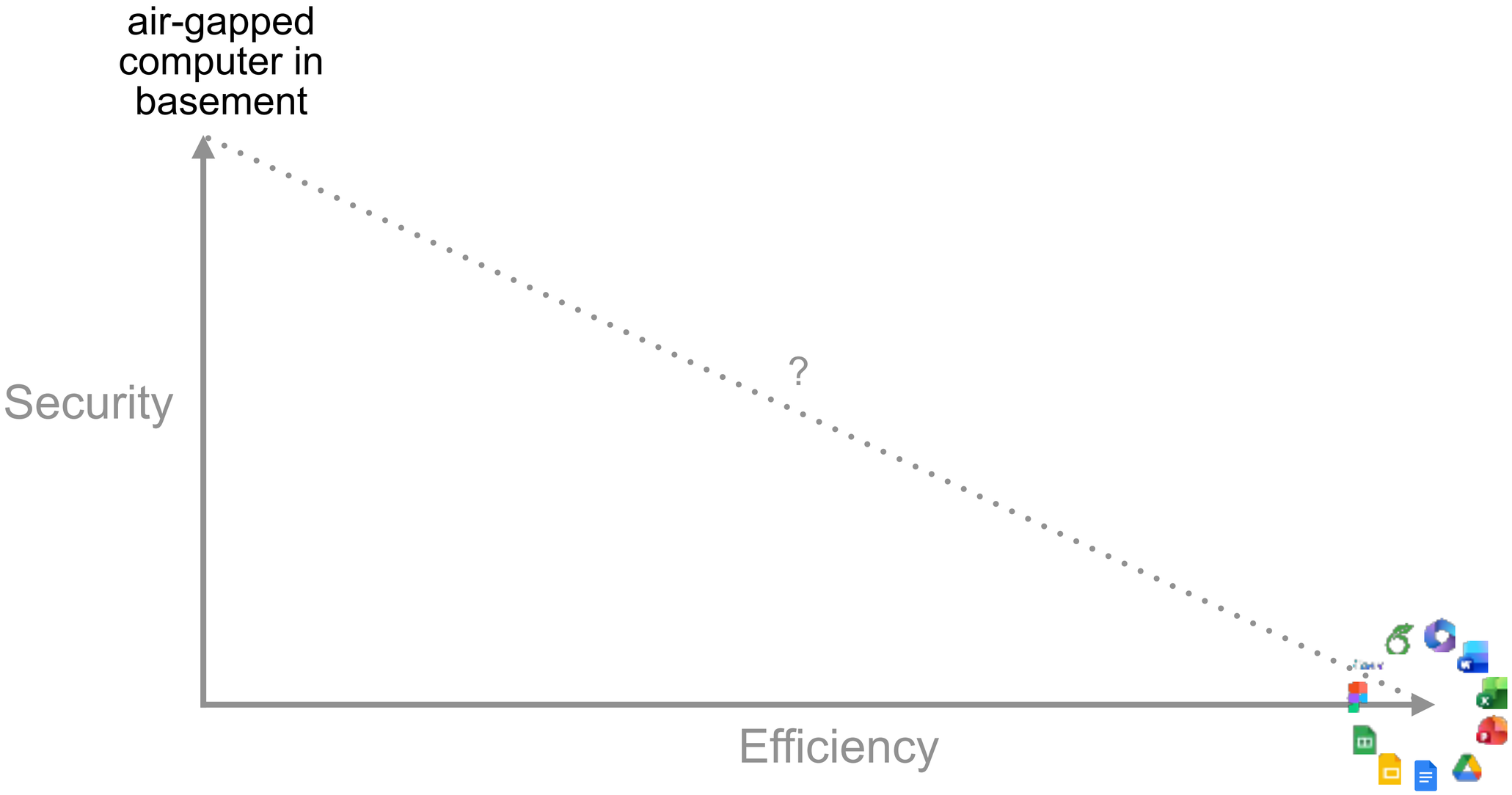
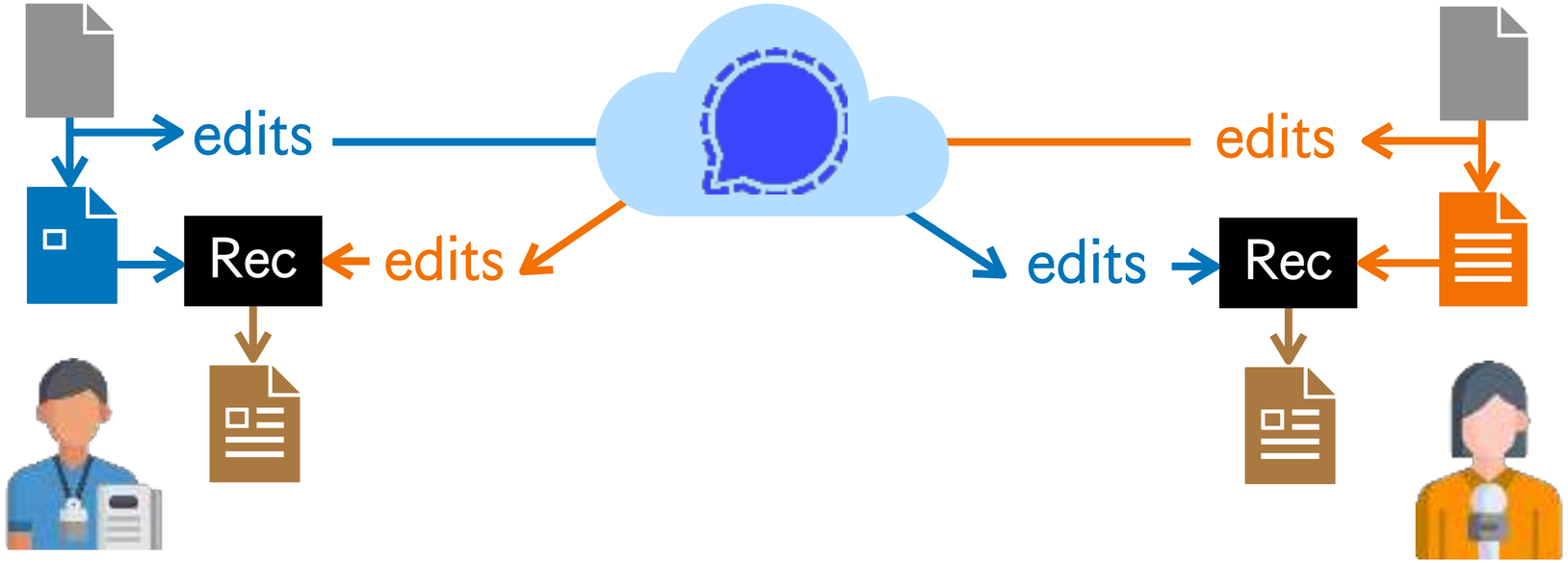


Evaluation setup

Edit: apply locally, send encrypted edit, reconcile



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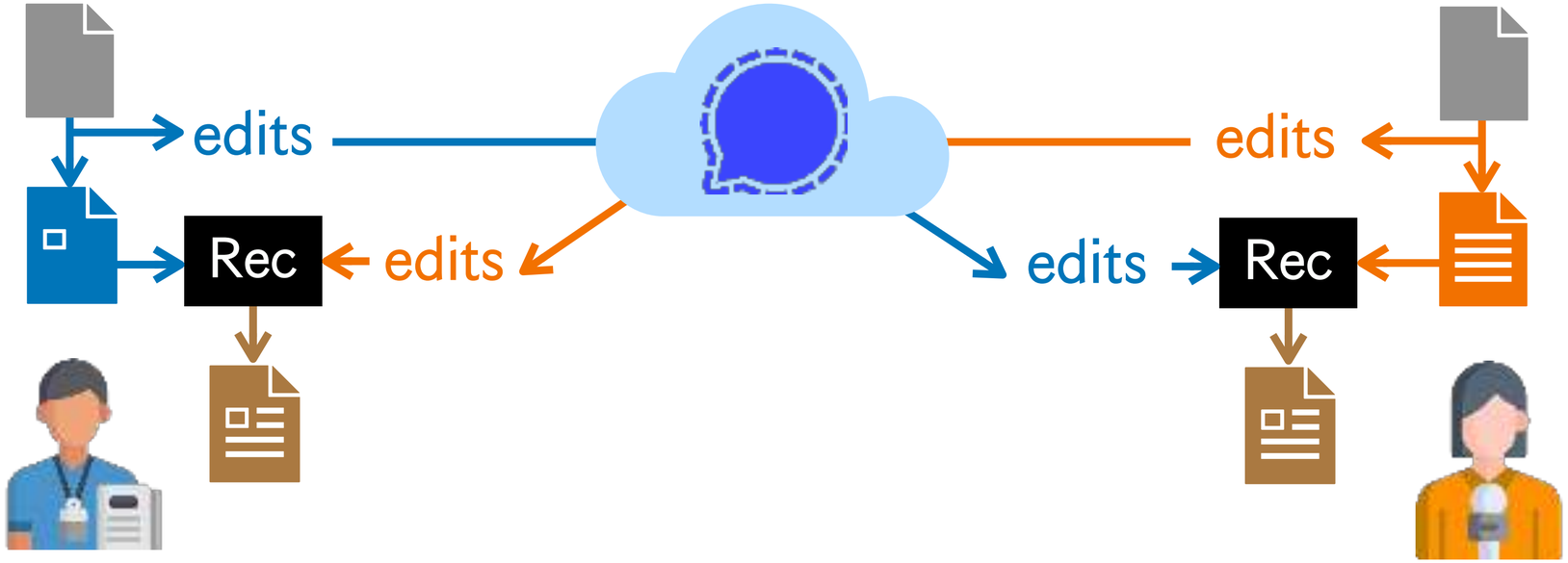

types at 40—60 wpm

types at 30—120 wpm

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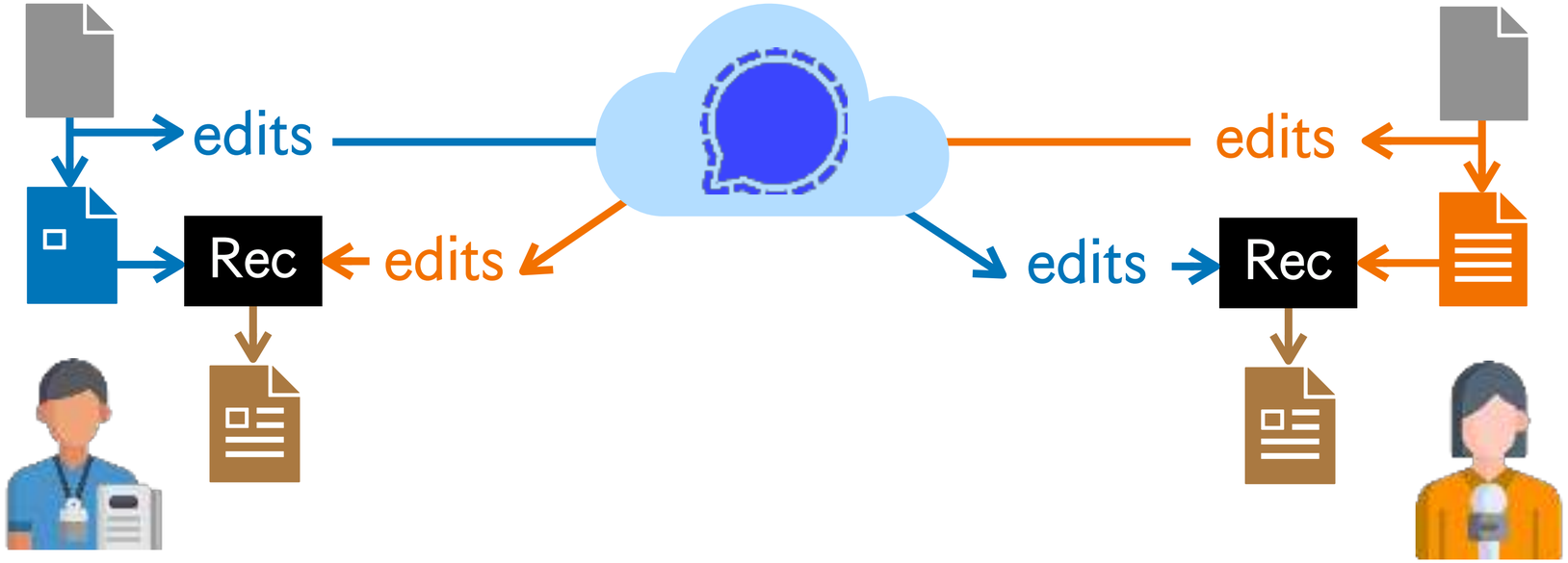

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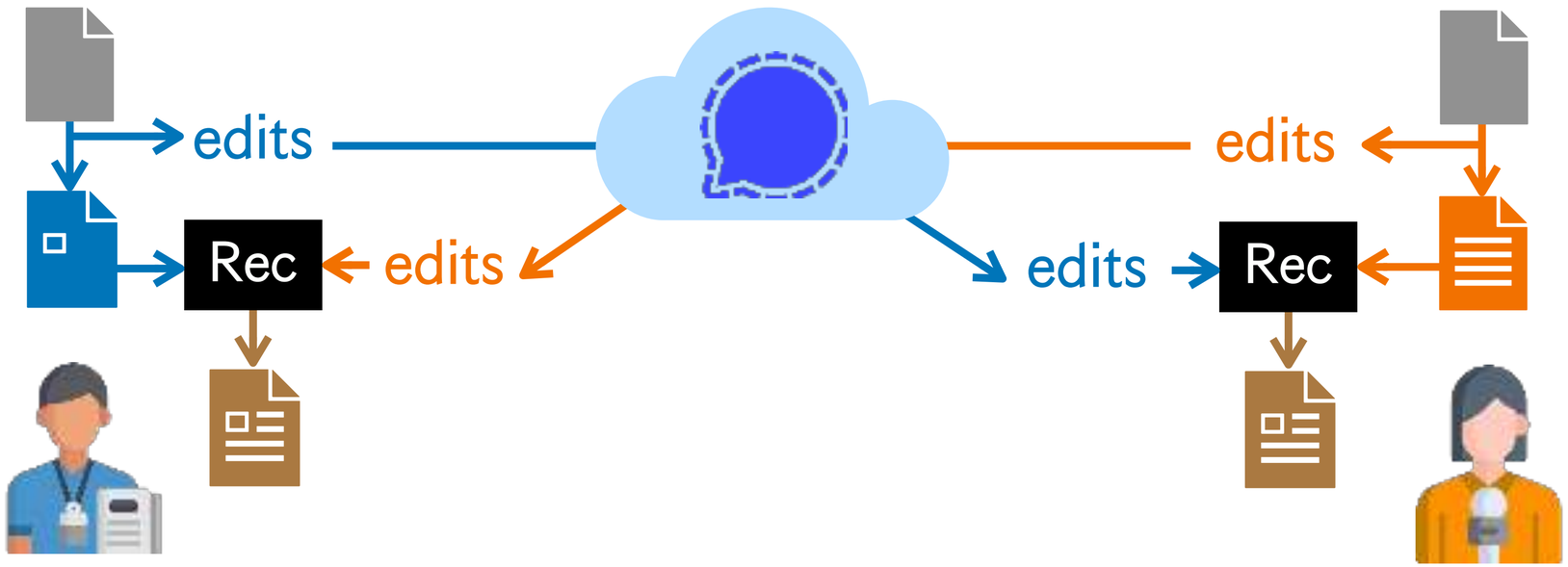

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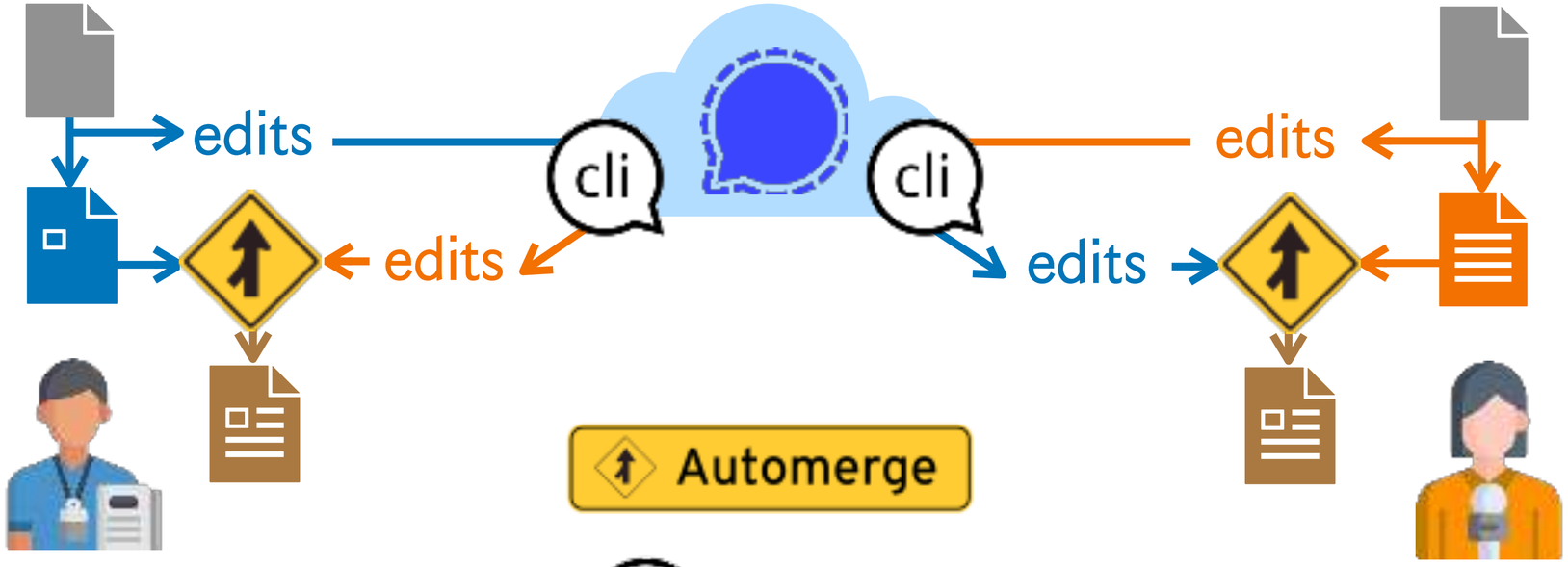

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cli signal-cli

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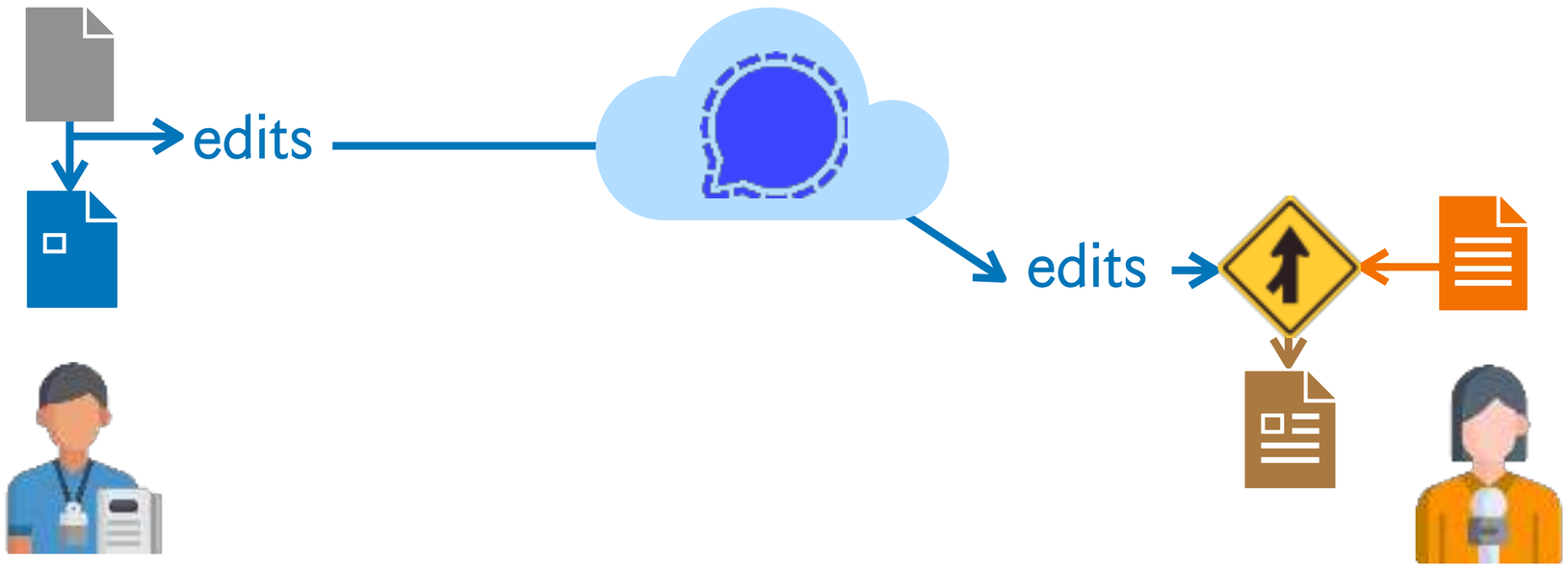
Synchronous editing



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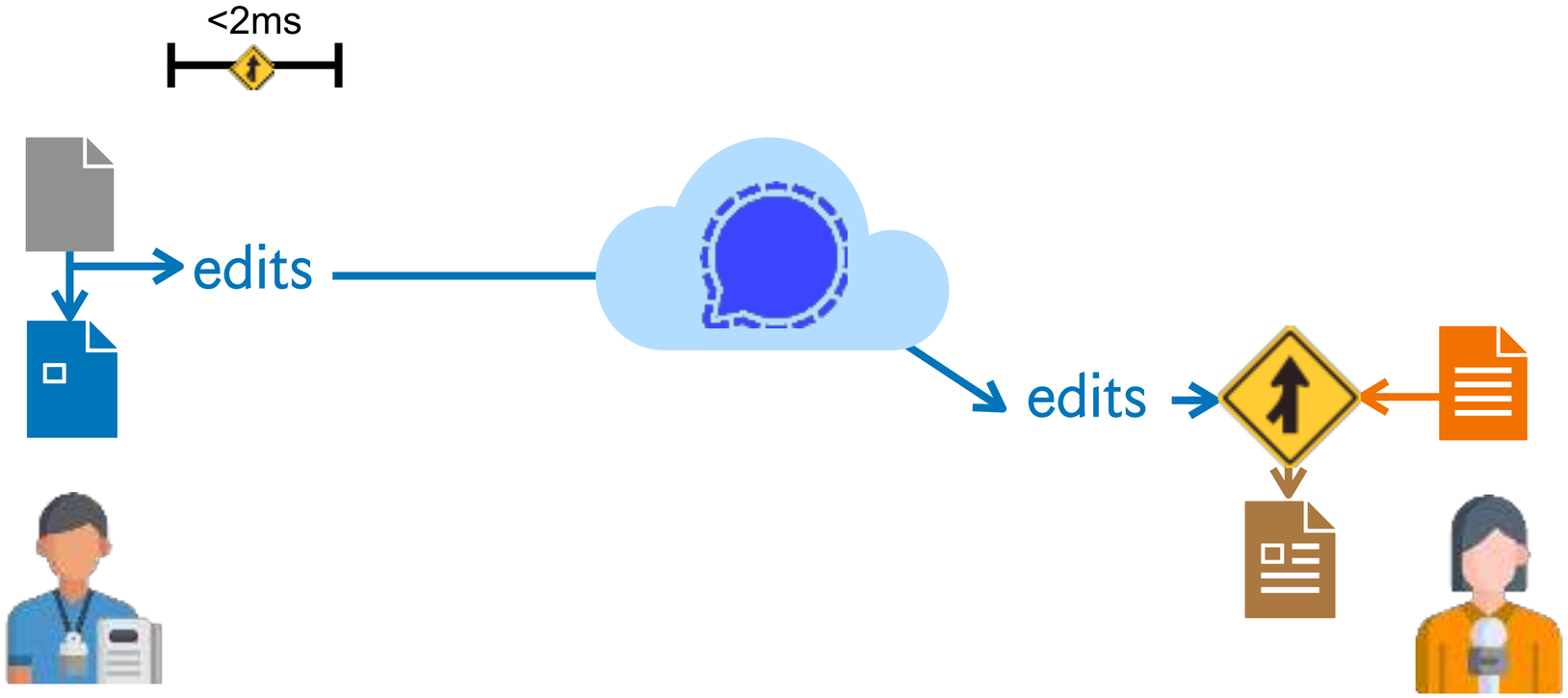

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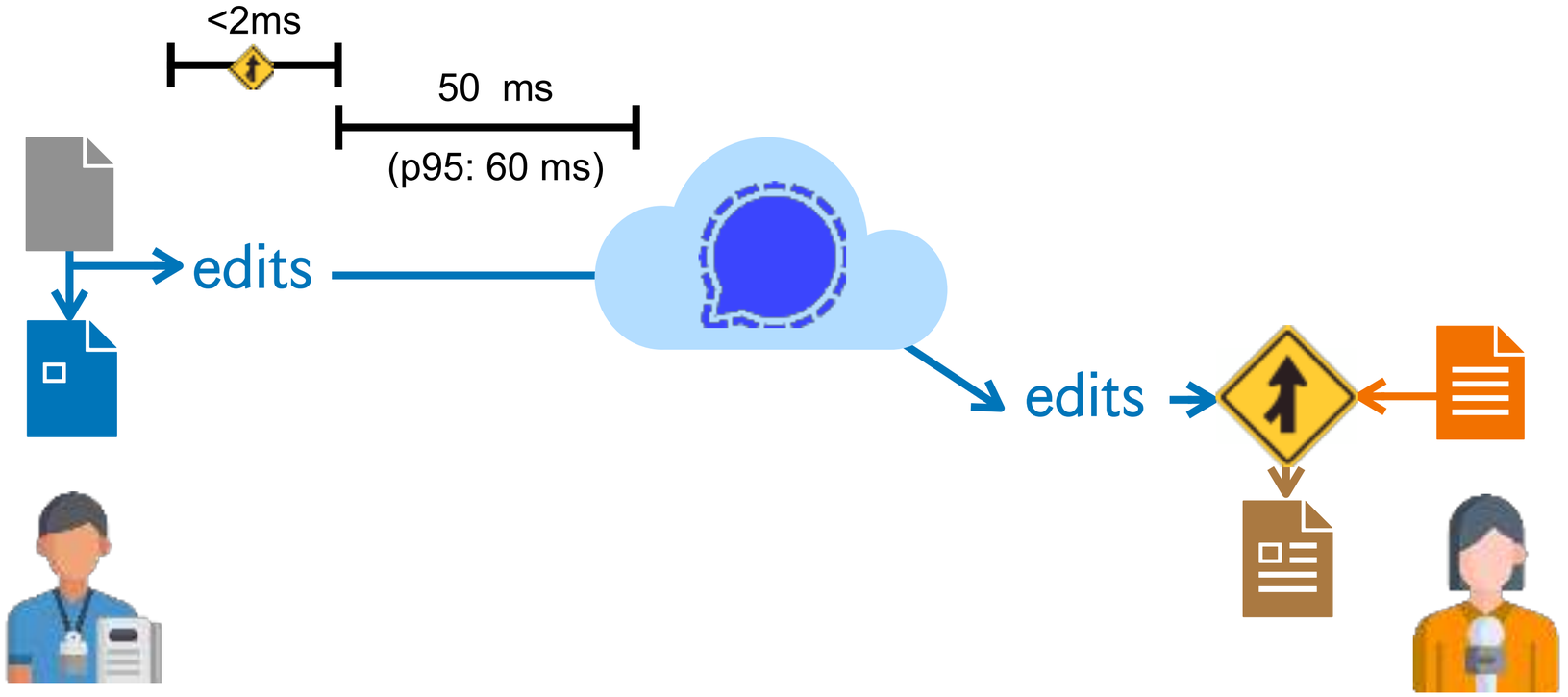

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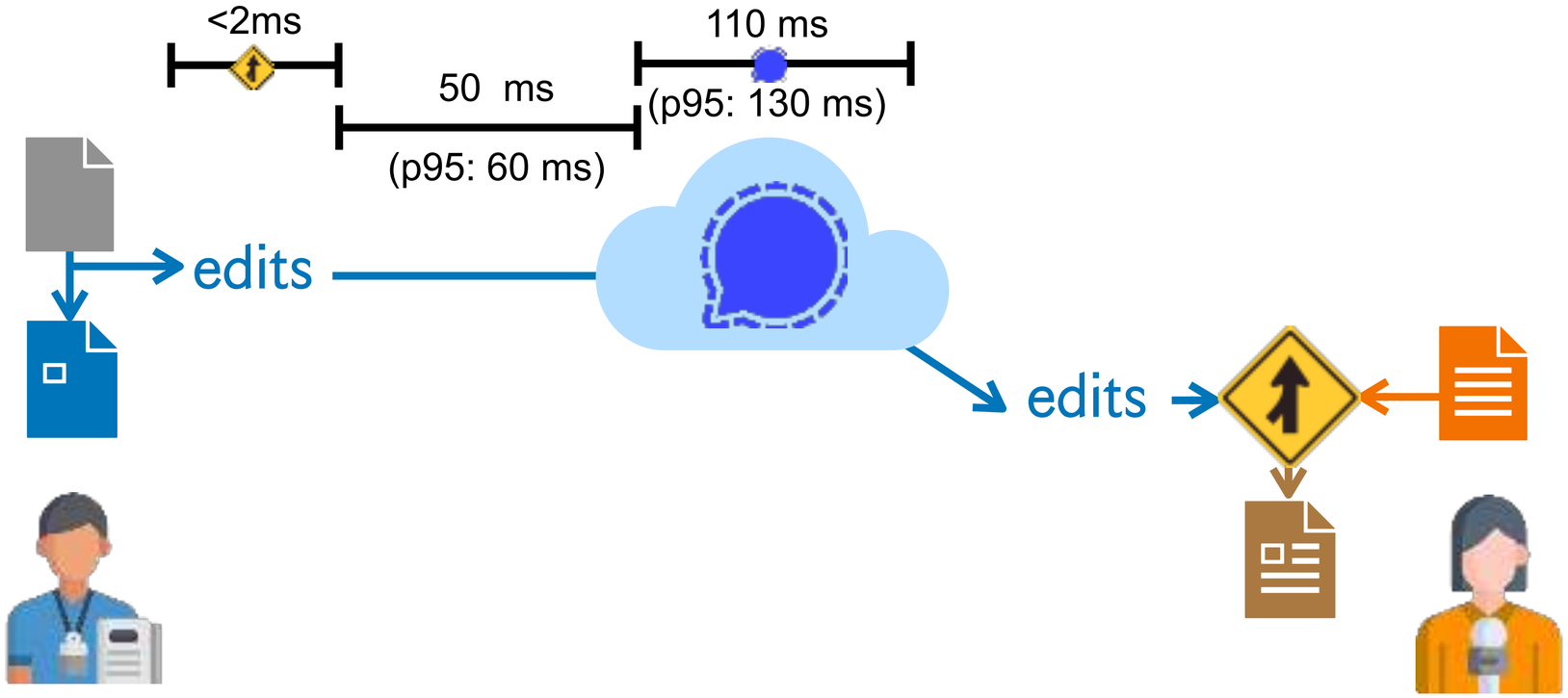

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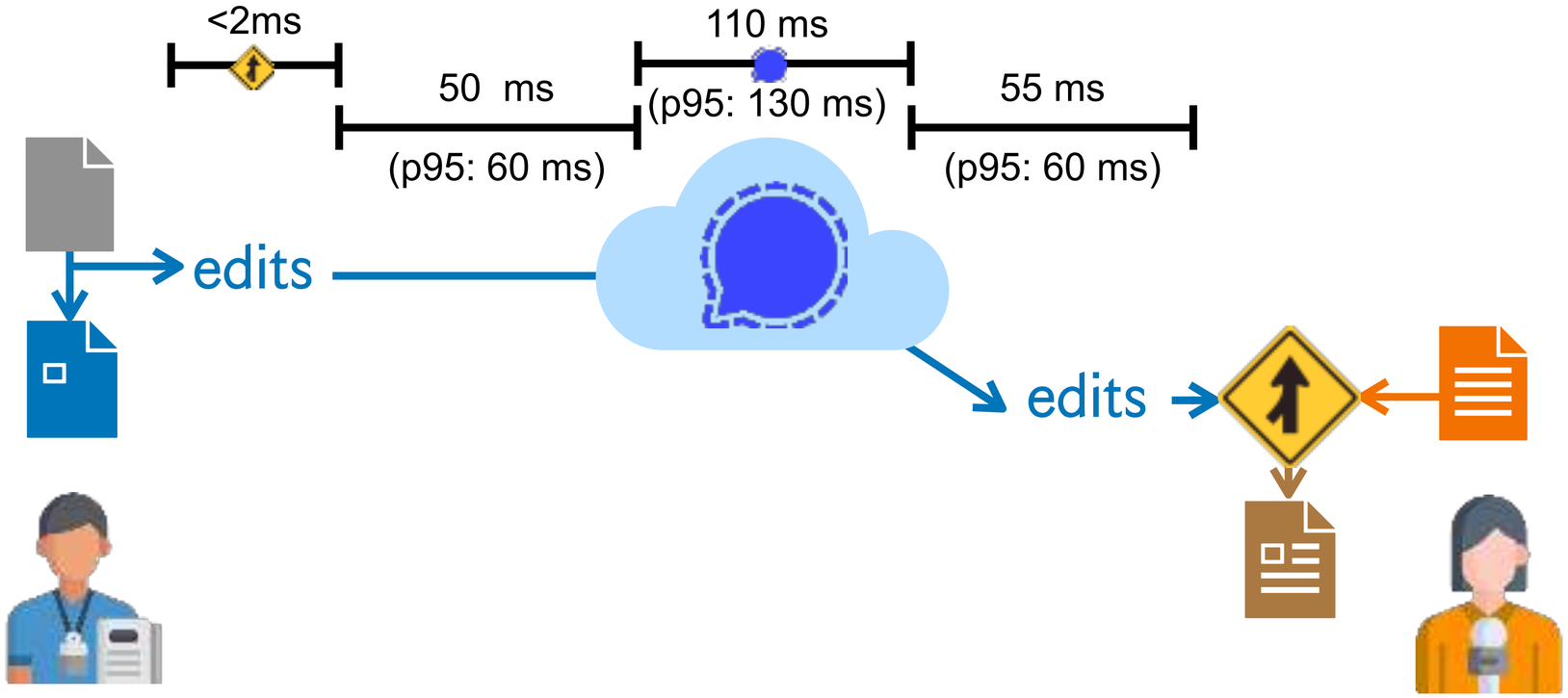
types at 30—120 wpm

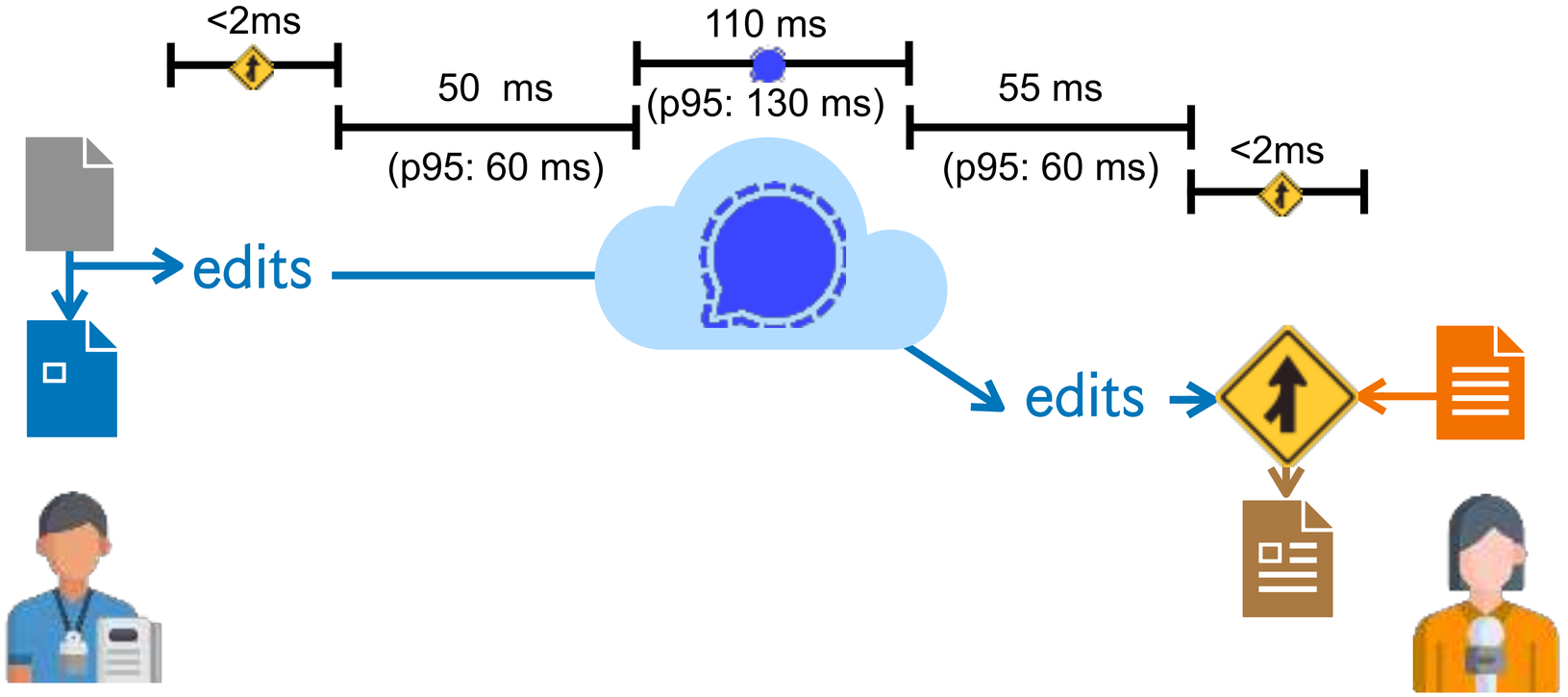
Synchronous editing



types at 40—60 wpm

types at 30—120 wpm

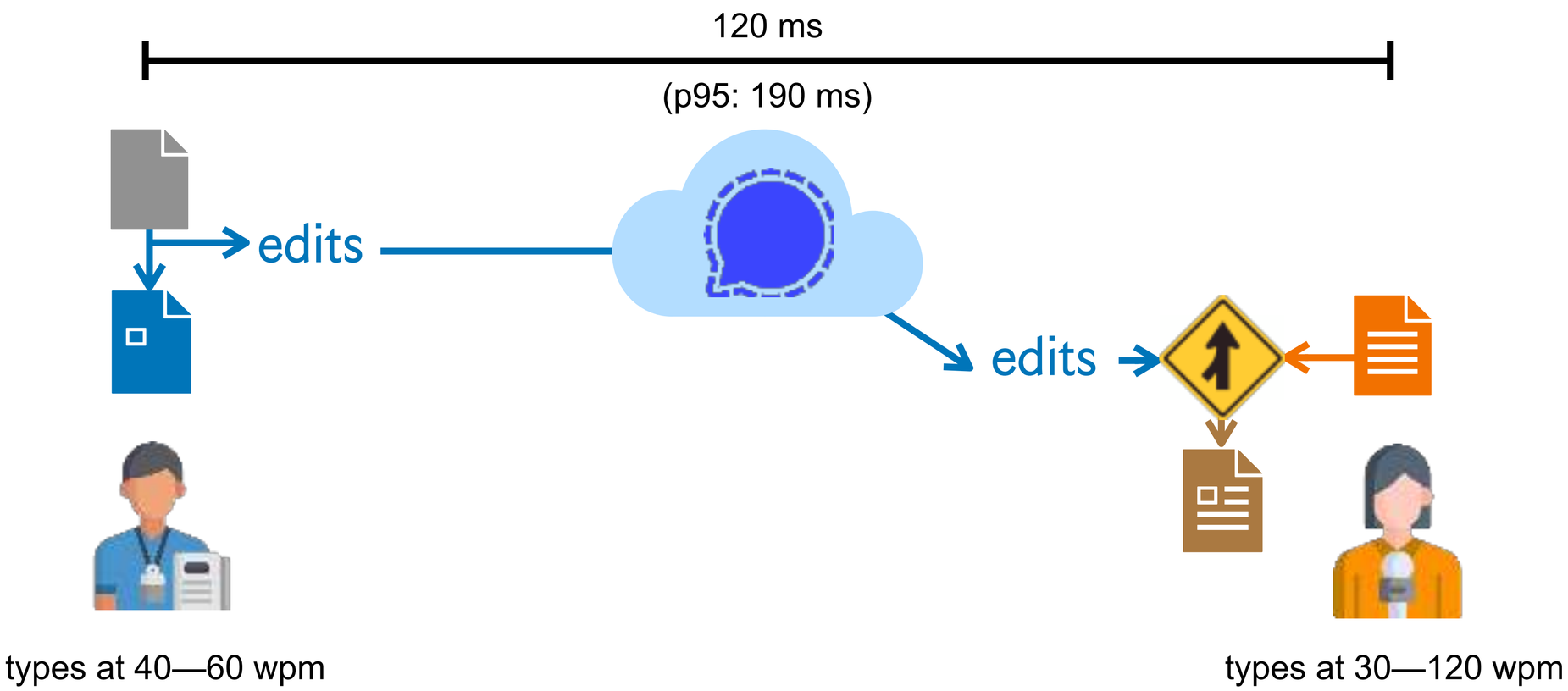
Synchronous editing



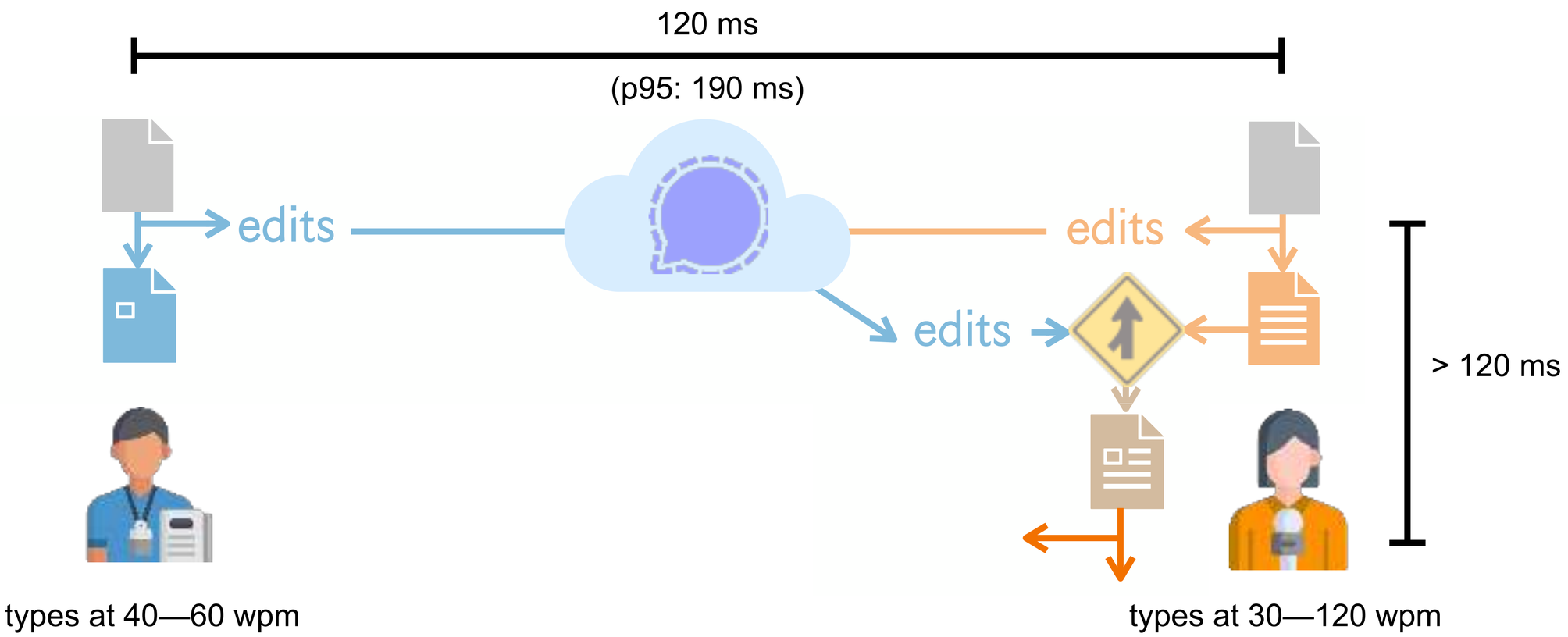
types at 40—60 wpm

types at 30—120 wpm

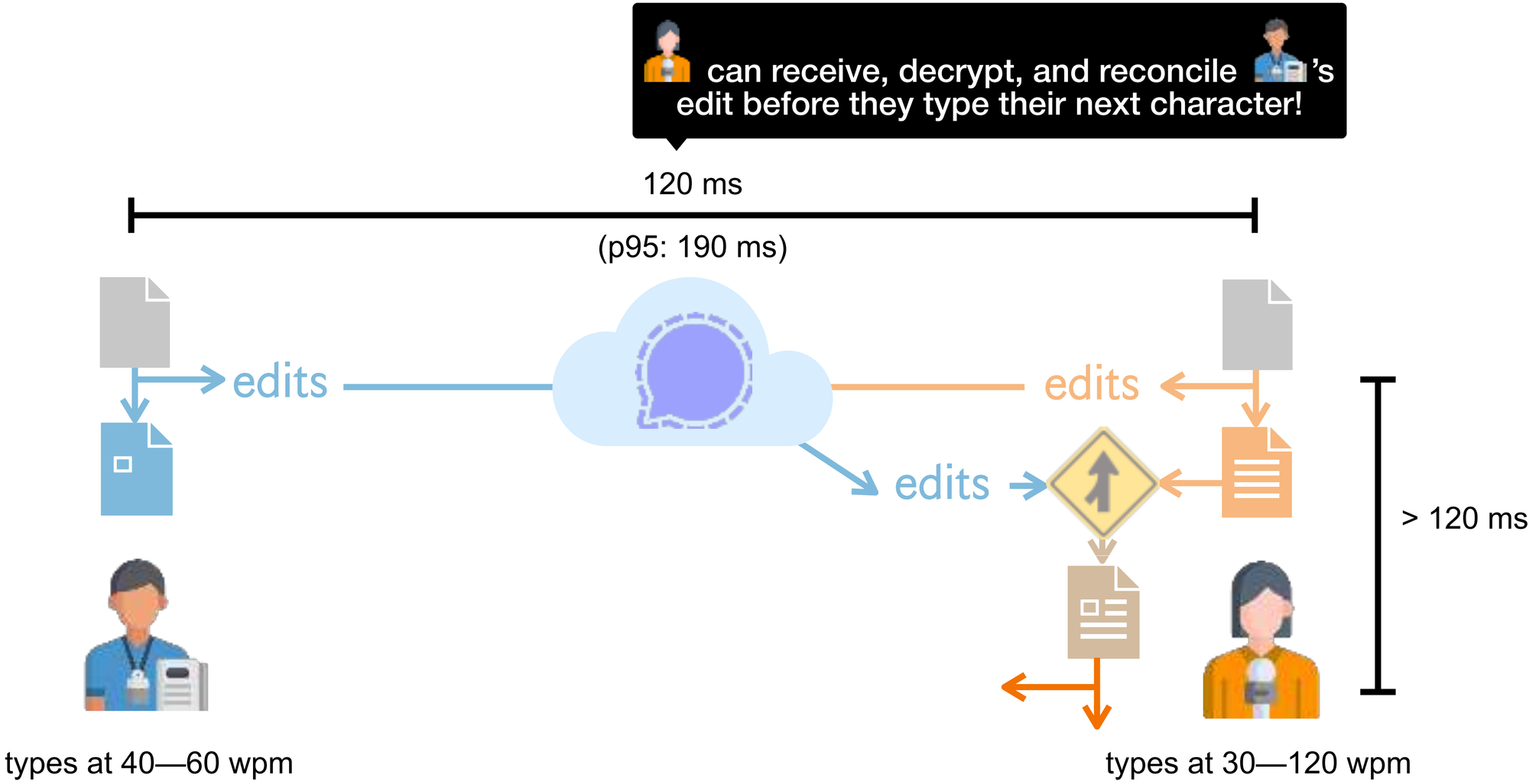
Synchronous editing



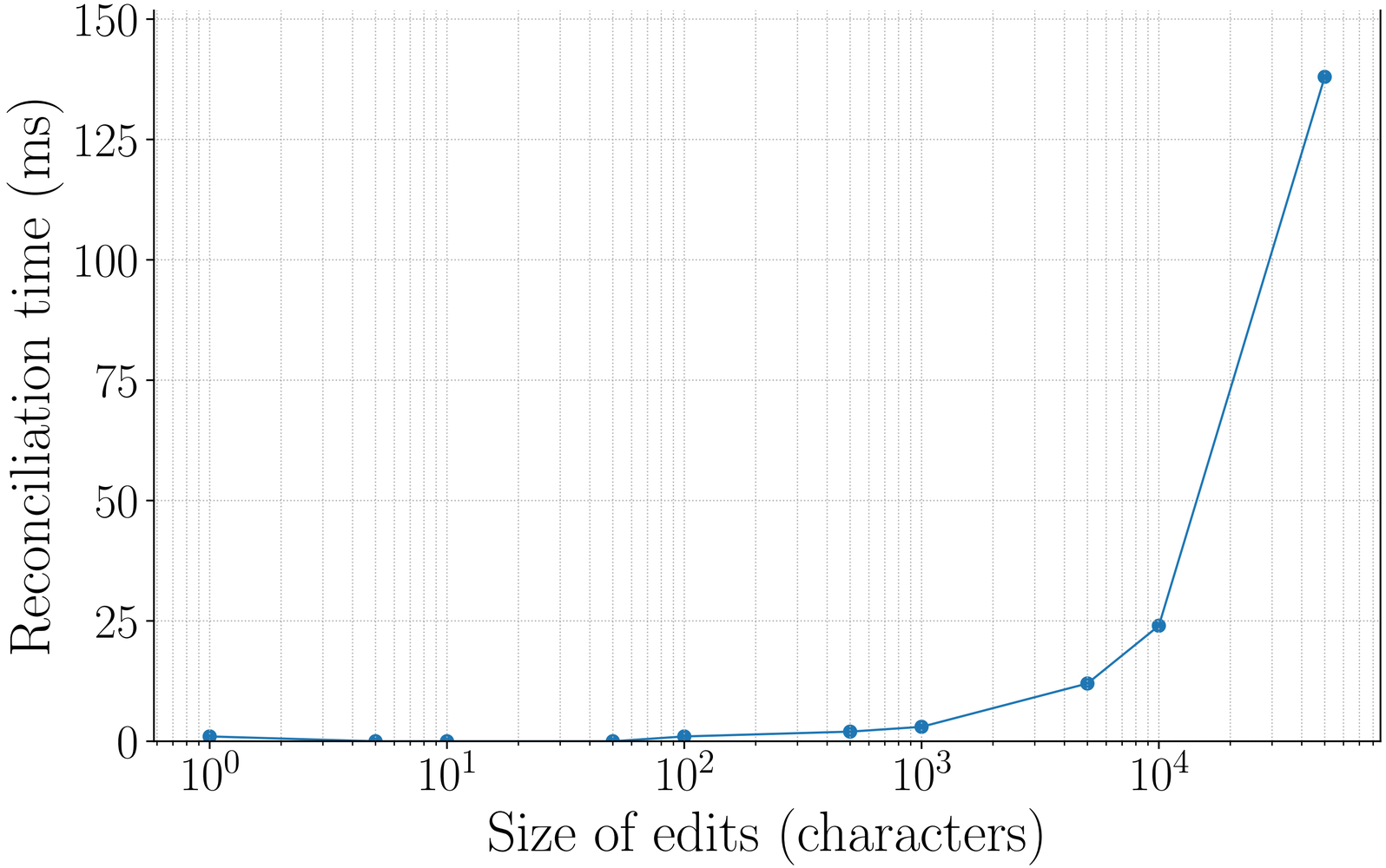
Synchronous editing



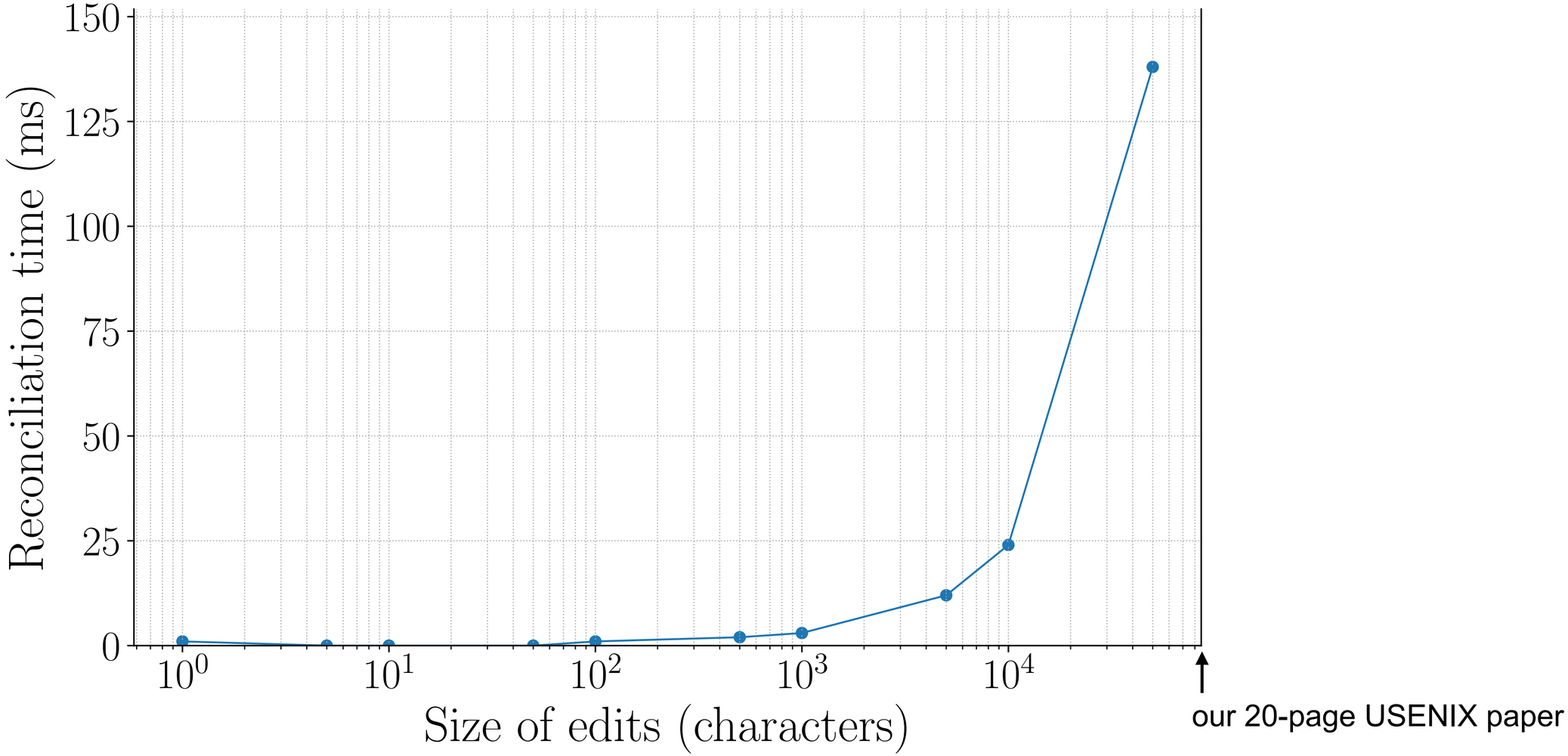
Synchronous editing



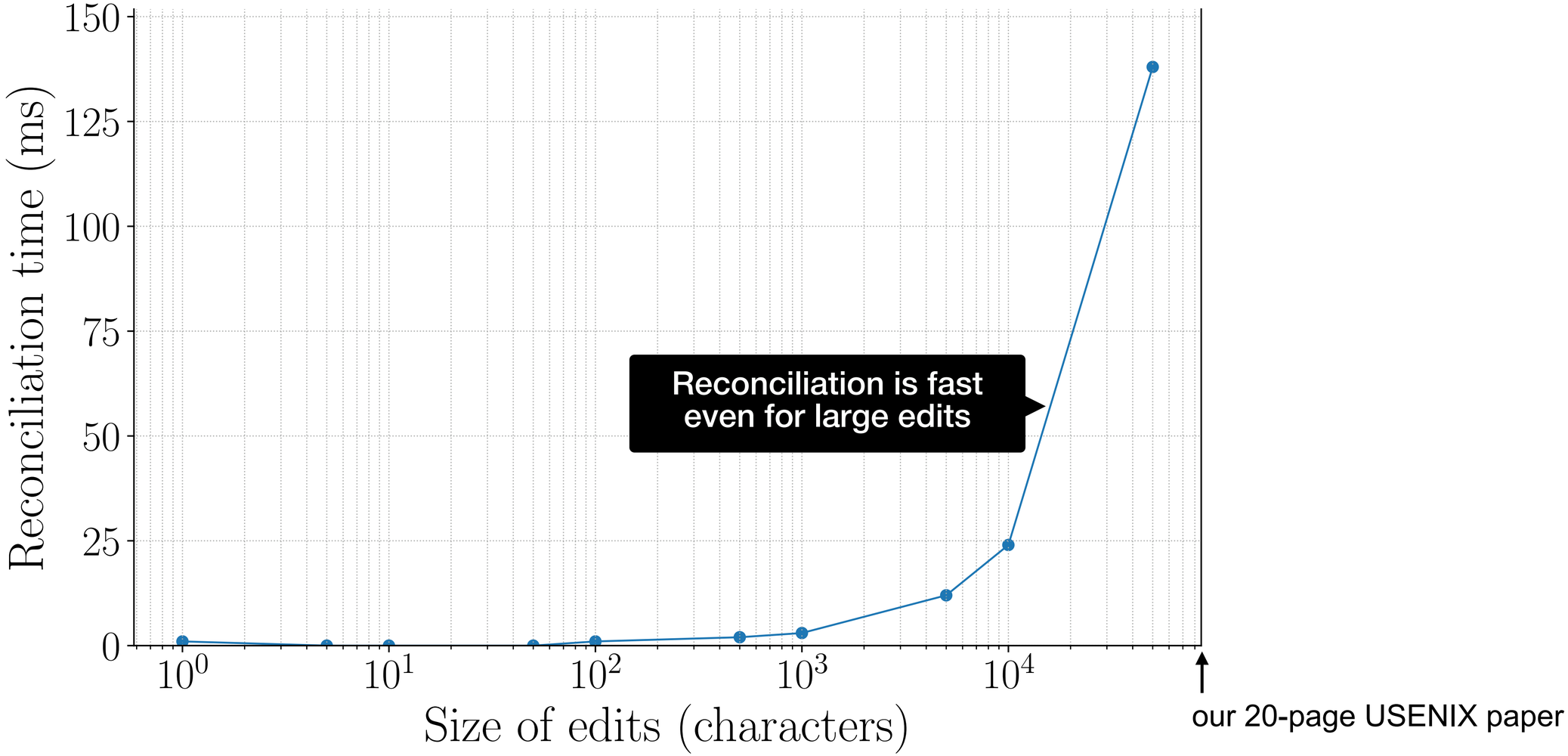
Large edits


150
125
100
75
50
25
0
Reconciliation time (ms)
10^0
10^1
10^2
10^3
10^4
Size of edits (characters)

Large edits


150
125
100
75
50
25
0
Reconciliation time (ms)
10^0
10^1
10^2
10^3
10^4
Size of edits (characters)
our 20-page USENIX paper

Large edits


150
125
100
75
50
25
0
Reconciliation time (ms)
10^0
10^1
10^2
10^3
10^4
Size of edits (characters)
Reconciliation is fast even for large edits
our 20-page USENIX paper

air-gapped computer in basement

collaborative docs from secure group messaging

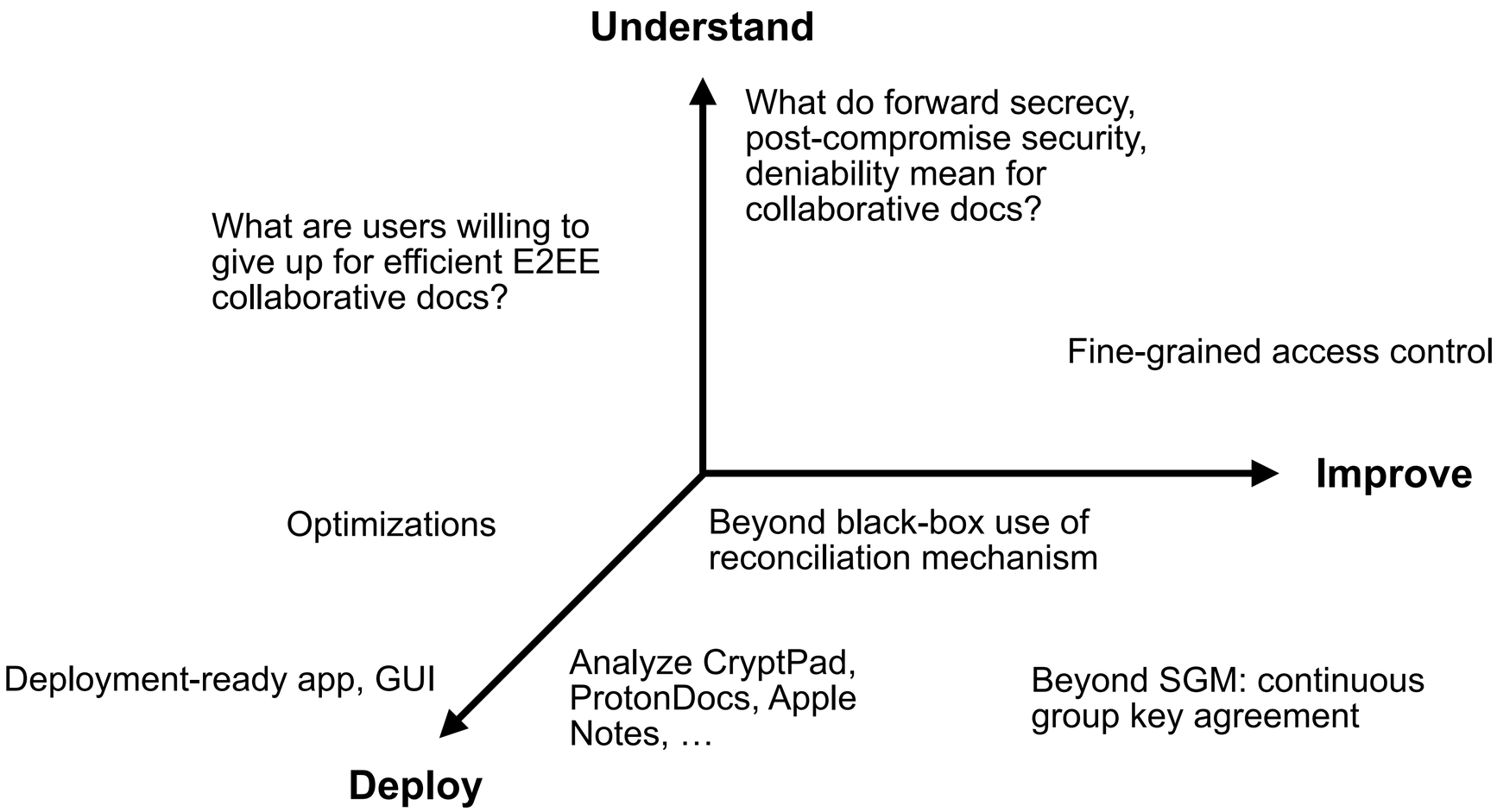
Security

Efficiency

Future Work

Why have secure collaborative documents not followed the path of secure messaging?

Future Work


Understand
What do forward secrecy, post-compromise security, deniability mean for collaborative docs?
What are users willing to give up for efficient E2EE collaborative docs?
Fine-grained access control
Improve
Optimizations
Beyond black-box use of reconciliation mechanism
Deployment-ready app, GUI
Analyze CryptPad, ProtonDocs, Apple Notes, …
Beyond SGM: continuous group key agreement
Deploy

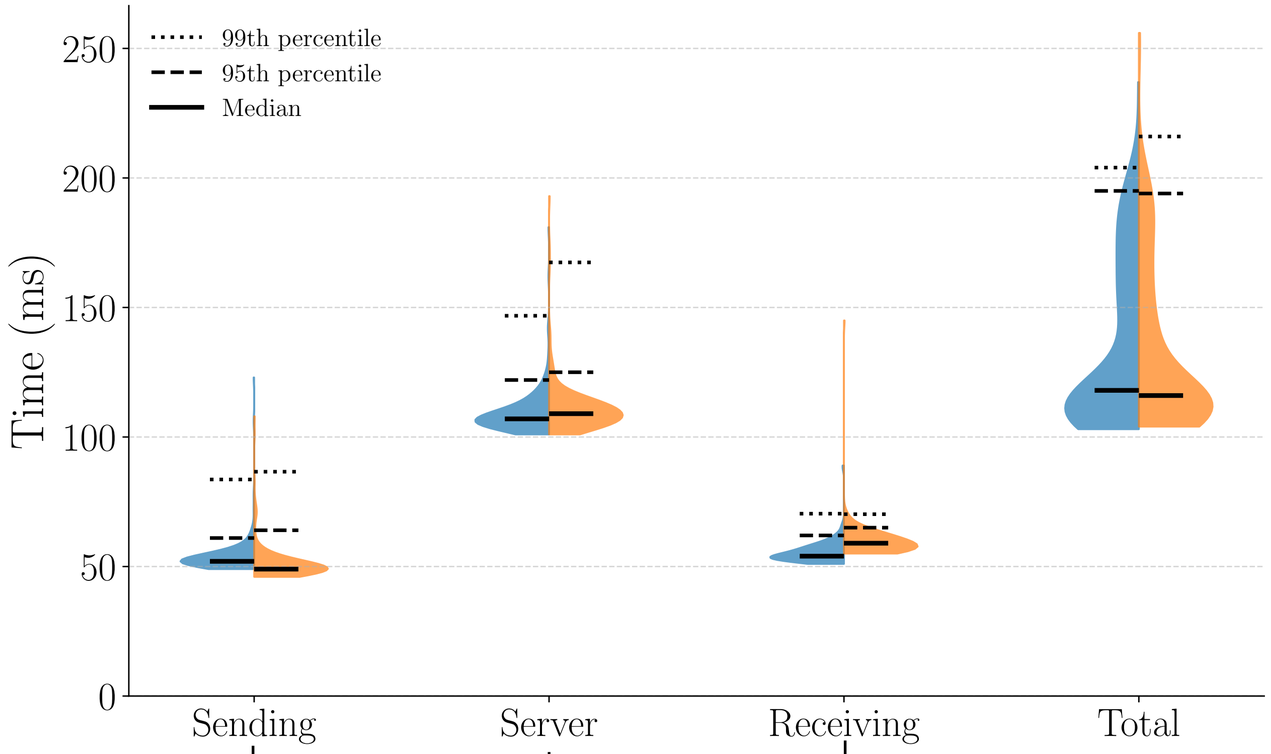
End-to-End Encrypted Collaborative Documents for Investigative Journalists

Christian Knabenhans EPFL
Zayd Maradni MPI-SP
Carmela Troncoso EPFL, MPI-SP

With thanks to
Investigative Journalists ICIJ
Alastair Beresford Cambridge
Daniel Hugenroth Cambridge, The Guardian
Martin Kleppmann Cambridge

To appear at USENIX Security'26

Synchronous editing

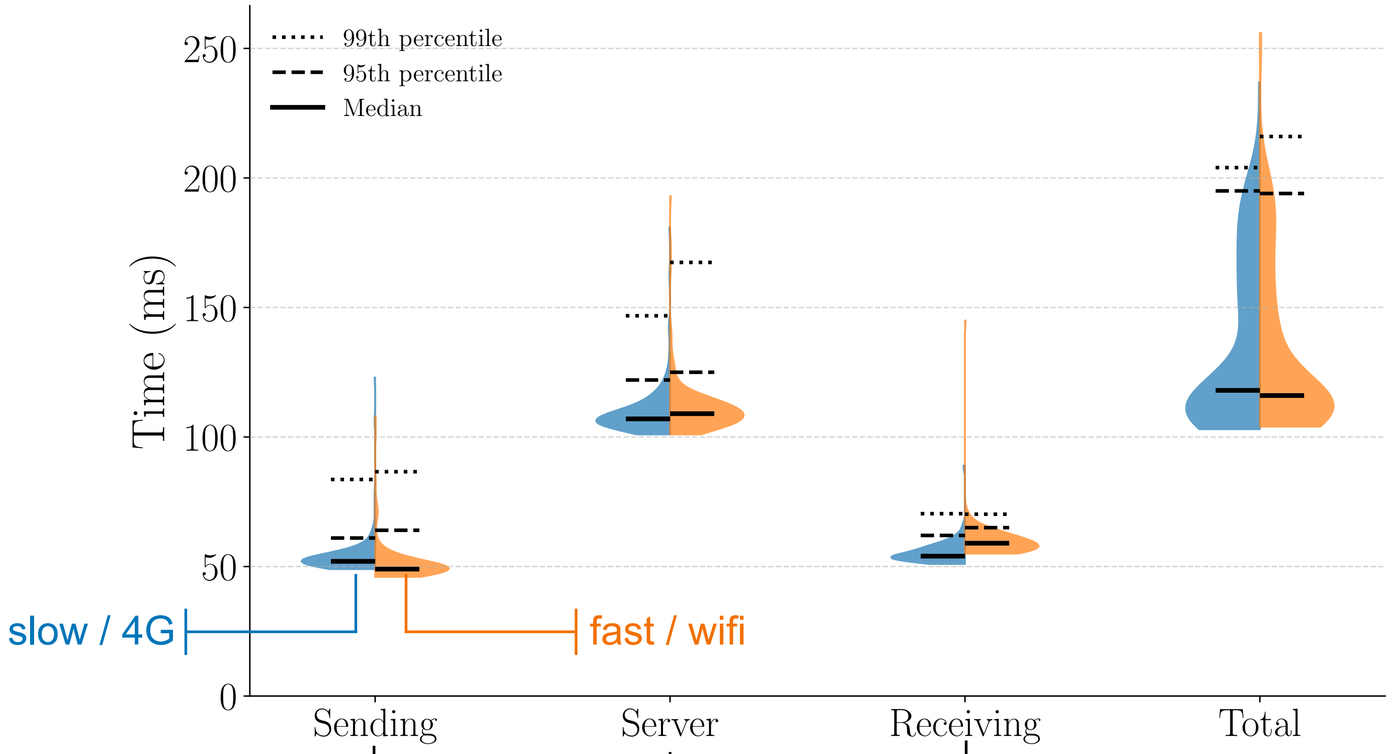


Time from sending to server ack

Time from server ack to server send

Time from server send to reception

Synchronous editing

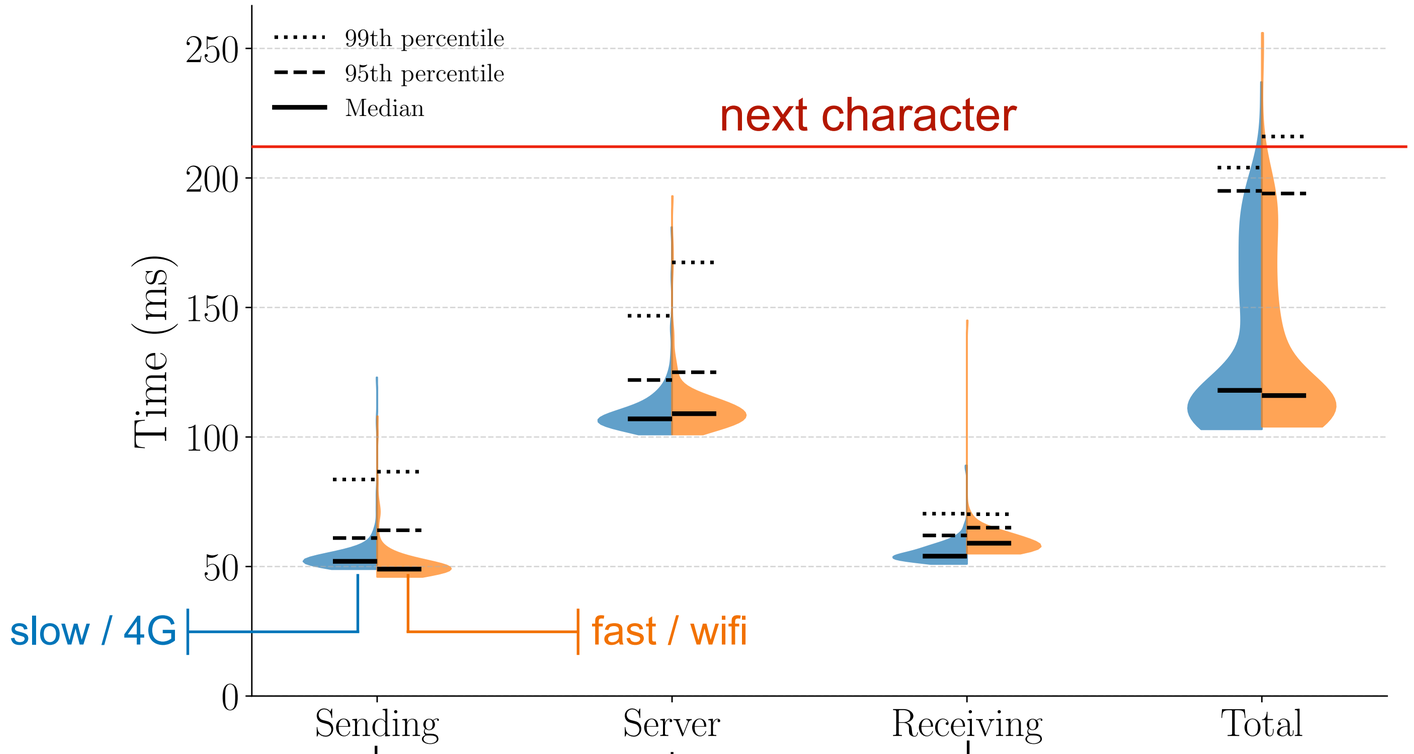


Time from sending to server ack

Time from server ack to server send

Time from server send to reception

Synchronous editing

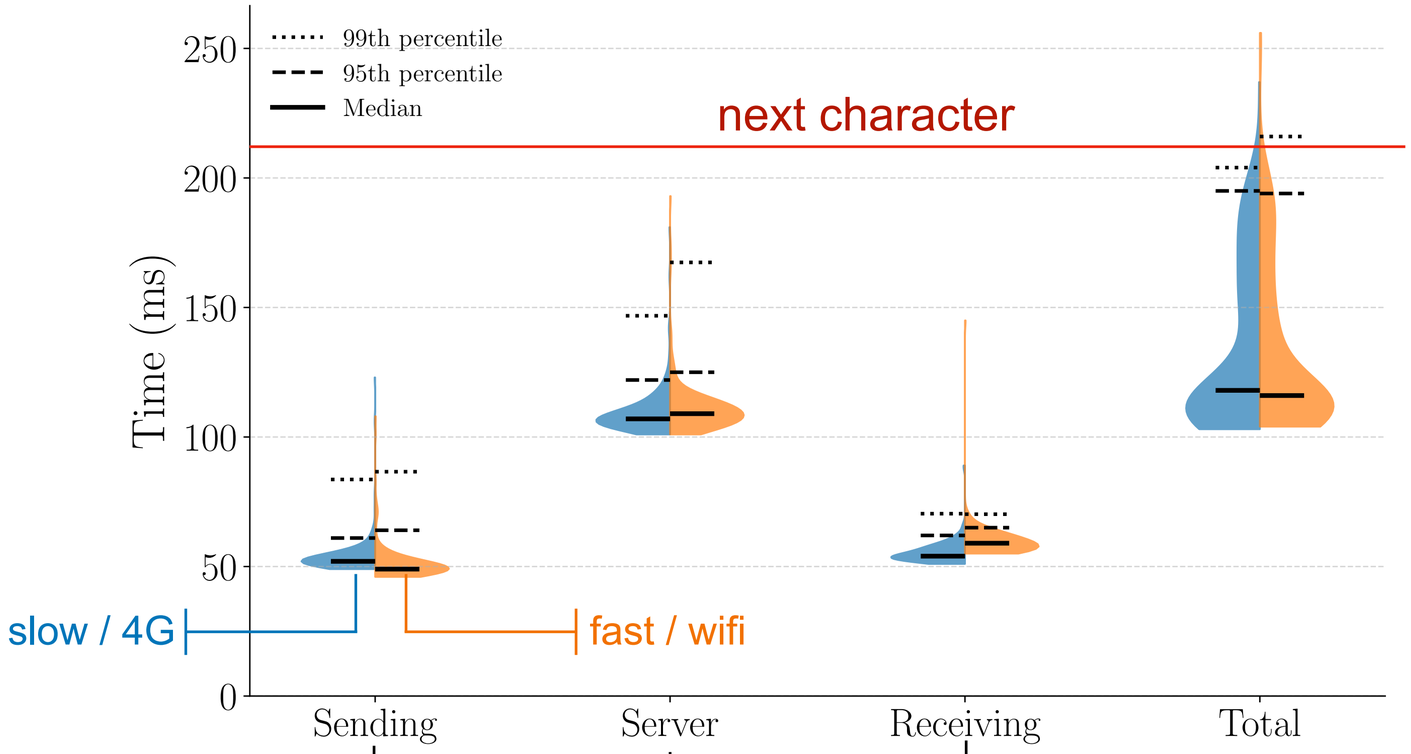


Time from sending to server ack

Time from server ack to server send

Time from server send to reception

Synchronous editing



Reconciliation time is <2ms for single-character edits

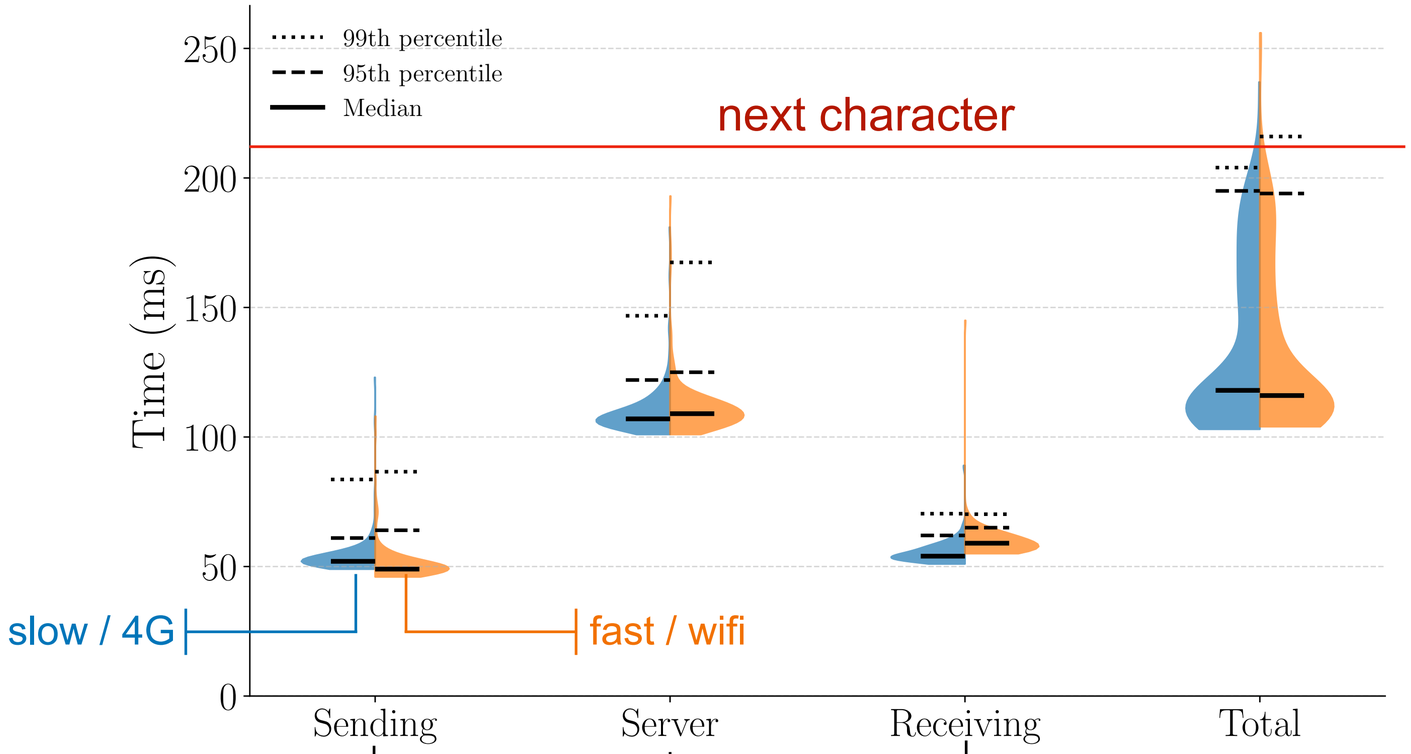
Time from sending to server ack

Time from server ack to server send

Time from server send to reception

Synchronous editing

User 2 can receive, decrypt, and reconcile User 1's edit before they type their next character!



Reconciliation time is <2ms for single-character edits

Time from sending to server ack

Time from server ack to server send

Time from server send to reception

Asynchronous editing

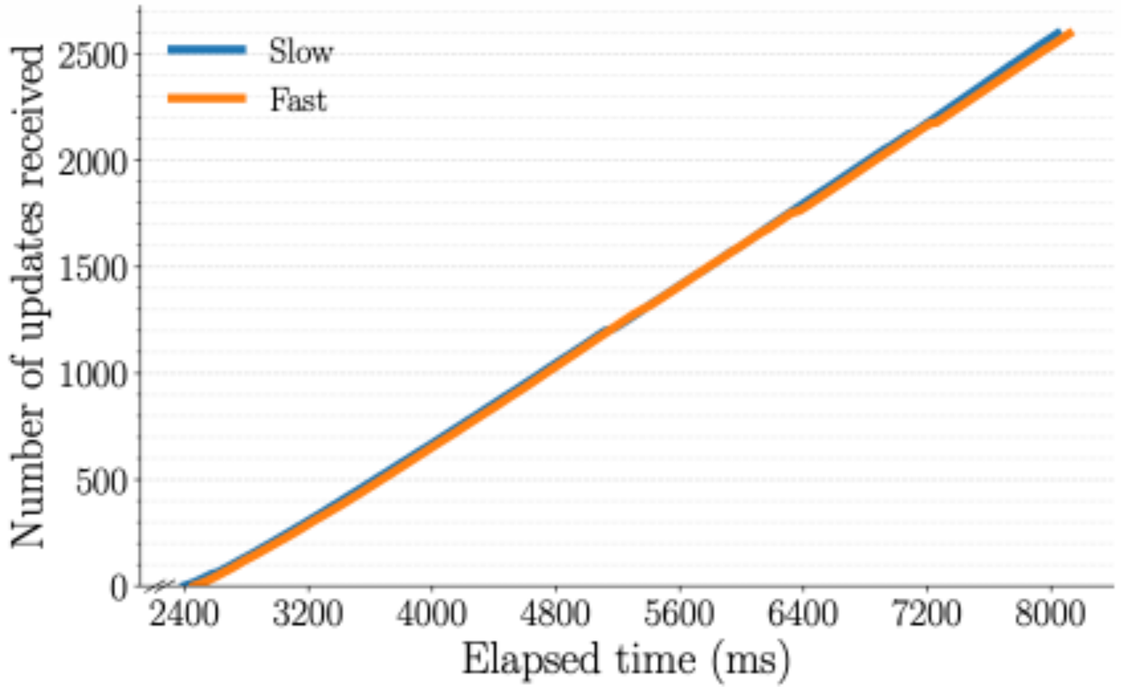


Figure 7: Received edits as a function of time since launch, for the `slow` and `fast` network configurations. After 2.4 s to launch a new instance of the `signal-cli` client, users can process roughly 460 single-character edits per second, even with a slow connection.

Asynchronous editing

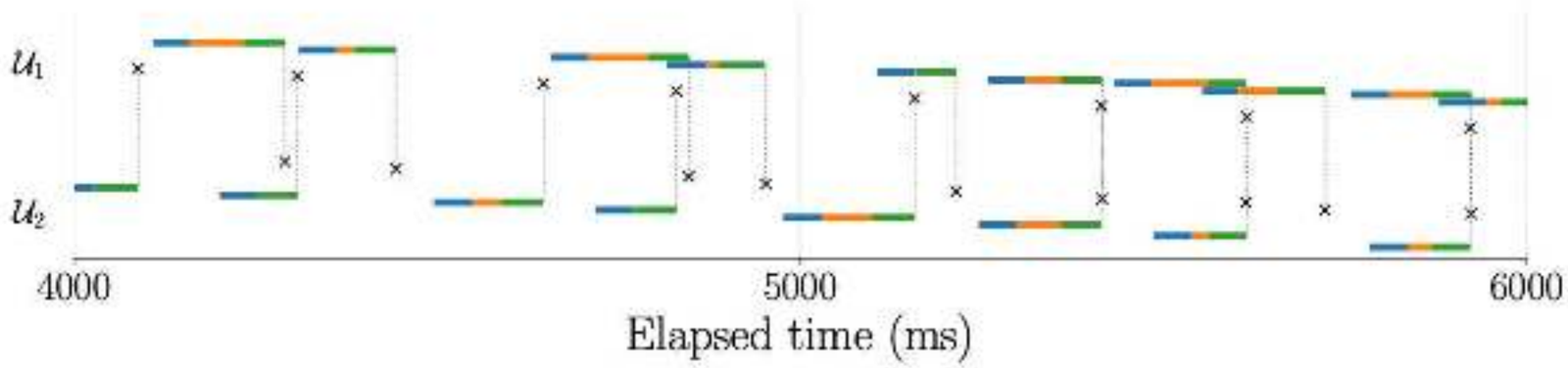
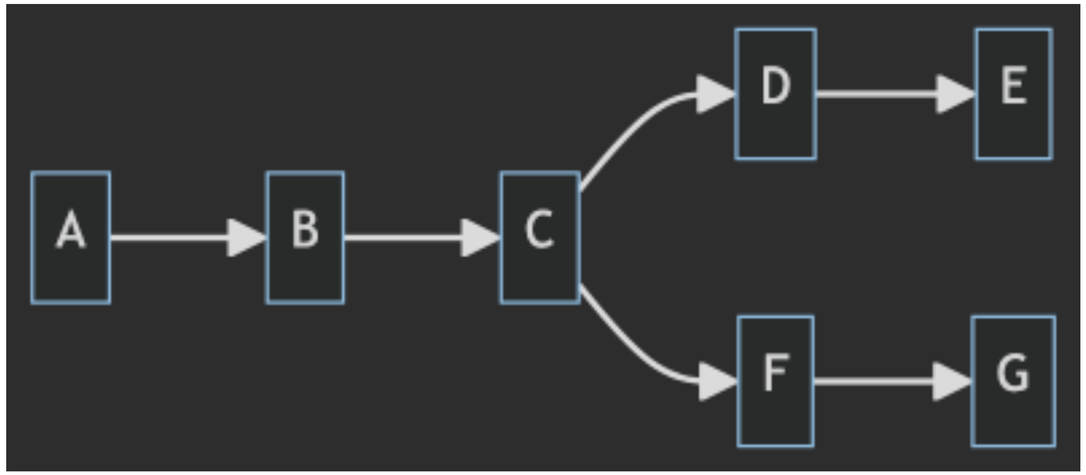
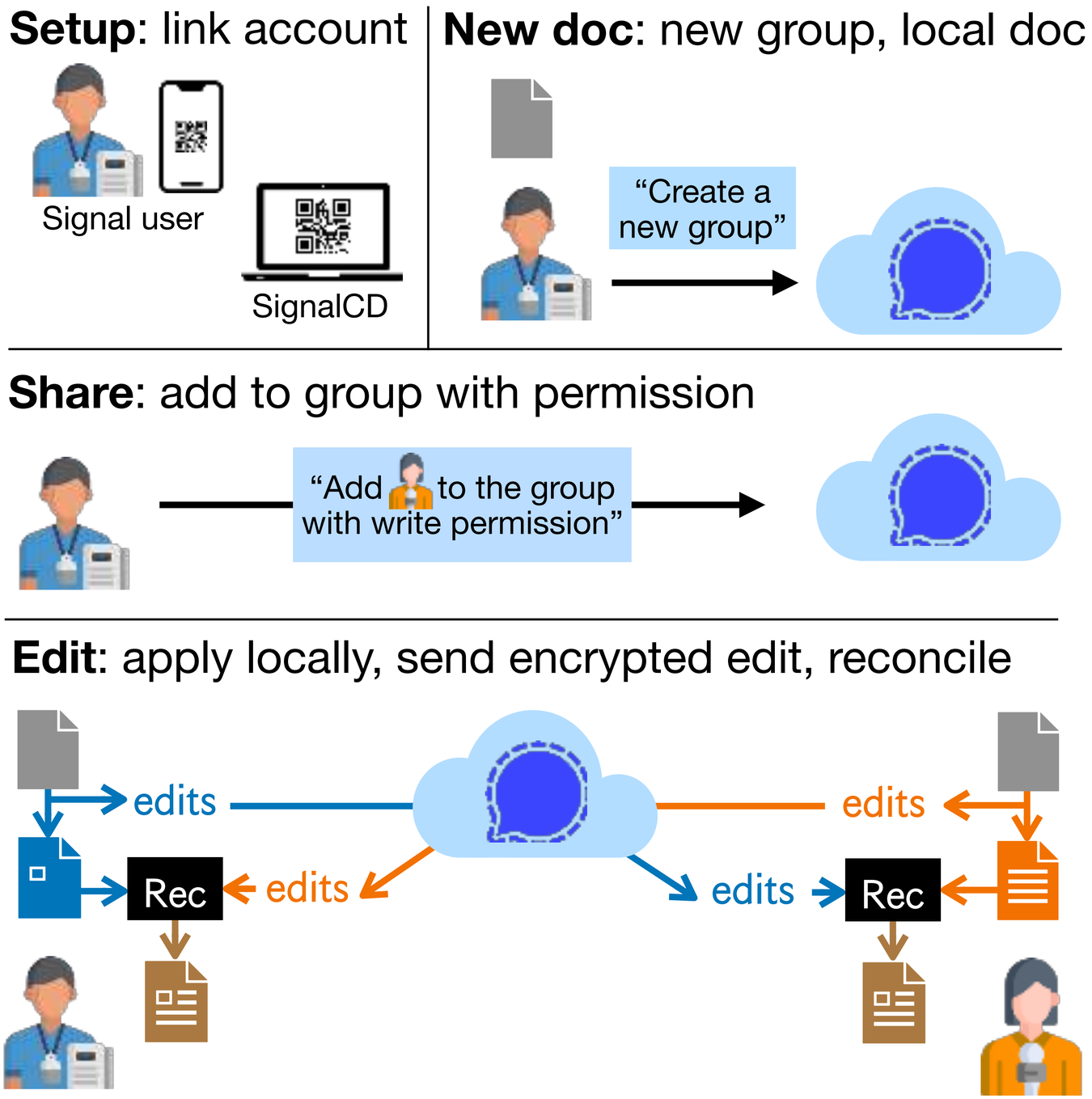


Figure 10: Staggered latencies for concurrent edits over a two-second interval (part of a longer editing session). Horizontal lines show the time from encrypting and dispatching the sending request until reception by the Signal server (blue), followed by the time until the Signal server delivers the message to the recipient (orange), followed by the time until the recipient's client application receives the message (green); the black cross marks the point at which the other user has applied the edit using Rec.

Automerge Example

- Document state has two concurrent edits (D,E) and (F,G) that need to be merged
- Non-conflicting edits: editing character at different keys, both deleting character etc.
- Conflicting edits: If both set the same key x to some value then randomly choose one value
- Note that “randomly choose” means “choose one arbitrarily, but in such a way that all nodes agree on the chosen value”.




Setup: link account
Signal user
SignalCD
New doc: new group, local doc
"Create a new group"
Share: add to group with permission
"Add to the group with write permission"
Edit: apply locally, send encrypted edit, reconcile
edits
Rec
edits
edits
edits
Rec